

Stakes and Investor Behaviors^{*}

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Abstract

We examine how stakes affect investor behaviors. In our unique setting, investors trade stocks in real accounts using their own money and simultaneously in a simulated setting. Our real-world within-investor estimation shows that investors exhibit stronger biases and perform worse in higher-stakes real accounts than in lower-stakes simulated accounts. Investors exhibit strong biases in both types of accounts, and the biases in both are strongly positively correlated. Such behavioral consistency suggests that low-stakes experiments are informative about real-world behaviors. Using additional account-level datasets, we demonstrate external validity by documenting a stronger (reverse) disposition effect on stocks (funds) with greater portfolio weights.

Keywords: stakes; behavioral biases; individual investors; household finance

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1. Introduction

Understanding the effects of stakes on human behaviors is of great interest to economists. First, understanding the stakes effect can shed light on the mechanisms that underlie the observed behaviors. For example, if a behavioral bias is driven largely by weak incentives, high stakes may increase the effort level such that the bias will be eliminated or significantly muted. Second, it also informs us whether insights gained in laboratories can be generalized to the real world. Economists have increasingly turned to the experimental method to understand human behaviors. Lab experiments enable investigators to conduct research under highly controlled conditions. Critics of the generalizability of lab studies have argued, though, that human behaviors may be sensitive to various factors that differ systematically between the lab and the real world (Thaler, 1986; Levitt and List, 2007; Benjamin, 2019). One of these factors is stakes.

This paper explores the effects of stakes on behaviors in a real-world stock market environment using data from a Chinese investor social network called Snowball. One unique feature of our setting is that many investors carry two types of stock accounts: real brokerage accounts in which investors trade using their own money and simulated accounts in which no real money is involved. Except for the fact that trades in simulated accounts do not involve real money, almost the same rules apply to these accounts as to real accounts. Such a unique setting enables us to construct a within-subject estimate of stakes effects. Using a sample of more than 4,000 investors over a three-year period, we compare the two types of accounts with reference to a standard set of prominent behavioral patterns that have been documented in the finance literature (Barberis and Thaler, 2003; Barber and Odean, 2013). As these patterns are typically considered suboptimal and most evidence shows that they are detrimental to investor performance (see Section 4.1), in this paper, following the finance literature, we often refer to them as biases.

Our setting is nearly ideal for studying the effects of stakes on investor behaviors. First, real accounts undoubtedly involve more substantial stakes than simulated accounts. Our conservative estimate suggests that the average real-account value is worth several times China's annual disposal income per capita. Second, critics believe that, in addition to being affected by high stakes, biases are likely to be eliminated or significantly muted by experience and competition in the real world, and it

is plausible that the cognitive processes underlying belief formation and revision operate differently in natural environments and in abstract lab settings (Thaler, 1986; Levitt and List, 2007; Benjamin, 2019). Third, although our sample is non-random, we emphasize that there is actual interest in studying populations that select endogenously into high-stakes decisions. A related criticism is that experimental subjects—typically students—may not accurately represent the relevant population in the real world. In other words, even if researchers can increase the stakes to the real-world level in labs, the effects observed in a lab may still differ from those in the real world as a result of other lab–field differences. Our real-world setting enables us to circumvent these criticisms.

Our headline result is surprising. In contrast to the conventional wisdom that higher stakes should reduce biases, we find that investors exhibit *stronger* behavioral biases in their real accounts than in their simulated accounts. In other words, investors do better when they care less. We observe in the real accounts a stronger disposition effect (a stronger tendency to sell stocks that earn capital gains than to sell those that incur capital losses), stronger lottery preferences (buying lottery-type stocks, which are lower-priced, more volatile, and more positively skewed than non-lottery stocks), greater extrapolation (buying stocks with high past returns), greater underdiversification (holding smaller sets of stocks in portfolios), and more active trading. Real accounts perform worse than their simulated accounts. The only behavioral pattern with respect to which we do not find a significant difference between real and simulated accounts involves local bias (the tendency to hold stocks in firms located relatively near where one lives).

Given our sample size and the advantage of the within-investor estimation strategy, we can estimate the effects of stakes precisely. Except for local bias, the differences in biases between real and simulated accounts are both statistically and economically significant. Even with respect to local bias, given the small standard error, we can confidently conclude that the difference between the two types of accounts, if any, is small. Consider extrapolation, underdiversification, and performance as examples. In real accounts, the average past-six-month return on purchased stocks is 15.60% higher than the market return, while it is 12.66% above the market return in simulated accounts. On average, investors hold 2.96 stocks in their real accounts and 4.36 stocks in their simulated accounts. In the two months following purchases, the sample stocks deliver average returns for real and simulated

accounts that are 0.66% and 0.15% lower, respectively, than returns on similar-sized stocks on the market.

These behaviors have been attributed to a range of cognitive biases (see Section 2 and Table 2). Although some heuristics and biases (e.g., prospect theory) can contribute to explaining more than one behavior (Barberis, 2013), the headline finding from this literature is that these behaviors are likely driven by distinct factors (Barberis and Thaler, 2003; Barberis, 2018). Consistent with this point, we find that these biases are only weakly correlated with each other. The fact that these biases are distinct and that we cover most of the widely studied investor behaviors in finance suggests the broad applicability of our findings, at least to stock markets.

The evidence also reveals systematic individual differences in the stakes effect. In other words, an investor for whom we find a greater real–simulated difference with respect to one bias typically exhibits greater real–simulated differences with respect to other biases. The findings of cross-investor heterogeneities in stakes effects naturally suggest that it would be sensible to explore individual characteristics to uncover the microfoundation underlying the importance of stakes. Our results show that stakes effects are weaker for investors who hold larger portfolios and those based in relatively developed regions. As our data largely lack individual characteristics, we await future research for results pertaining to this important question.

It is possible that stakes strengthen biases because, insofar as no real money is involved in simulated accounts, investors may trade randomly and hence show no systematic biases. We present several sets of evidence that are inconsistent with this conjecture. First, even though biases are generally weaker in simulated accounts than in real accounts, the biases we find in simulated accounts are strong in their own right. Second, we find strong positive correlations between investor biases in the two types of accounts, suggesting behavioral consistency even with no monetary stakes involved. These findings indicate that low-stakes experimental methods, although imperfect, can be informative about real-world human behaviors.

The real–simulated account comparison juxtaposes investor decisions involving high monetary stakes with counterpart decisions that involve no monetary stakes. Ideally, we hope to gather information that sheds light on investor behaviors at all stakes levels. To this end, we provide evidence using two additional account-level datasets—one from a US brokerage company and the

other from a Chinese brokerage company. In this setting, the variations in stakes involve distinct positions in an investor's portfolio. Positions that carry greater portfolio weights are thought to involve higher stakes. We find that investors exhibit a stronger disposition effect with regard to stock positions that carry greater portfolio weights. Positions where the portfolio weights vary all involve positive and, typically, high monetary stakes. Hence, the evidence we analyze from the two brokerage datasets shows that, with high-stakes investment decisions, increasing the stakes strengthens biases. Overall, with the caveat that portfolio weights are not determined randomly, these findings provide additional evidence supporting the proposition that higher stakes strengthen biases and also provide external validity supporting our analysis based on the real–simulated account comparison.

The psychology and experimental economics literature has proposed several mechanisms through which stakes may affect biases.⁴ One widely held view is that raising the stakes induces greater attention and effort, and greater effort can lead to more accurate translation of individuals' beliefs or preferences into their behaviors. The proposal that beliefs and preferences are elicited through financially incentivized designs is built on this view (Grether and Plott, 1979; Grether, 1980).⁵ The conventional wisdom that higher stakes should reduce biases assumes that greater attention and effort leads people to shift from using heuristics to engage in more deliberative processing, form less noisy expectations, and make better decisions (Stanovich and West, 2008; Gabaix, 2014, 2019; Imas, Kuhn, and Mironova, 2022).

There are at least two nonexclusive mechanisms through which higher stakes may instead strengthen biases in the financial context. The first mechanism challenges the assumption underlying the abovementioned conventional wisdom. The marginal effects of stakes can be negative if individuals carry incorrect mental models and greater attention leads to the more careful application of those incorrect models (Dawes, 1979; Dawes, Faust, and Meehl, 1989; Giglio et al., 2021; Hartzmark, Hirshman, and Imas, 2021). For example, if investors believe in price momentum, higher stakes can induce greater effort in locating stocks that have earned higher past returns, strengthening

⁴ We review the stakes literature more comprehensively in Section 2.

⁵ This mechanism relies on the assumption that decision-making is costly. When the cost of decision-making is zero, investors will always translate their beliefs and preferences perfectly into their behaviors and their actions will not differ regardless of the stakes. That decisions in the financial context are costly seems obvious because investors need to collect information, form beliefs, and perform utility-maximizing calculations.

extrapolation bias. Similar arguments also apply to cases involving intrinsic but nontraditional preferences, such as prospect theory, realization utility, cognitive dissonance, and sensation-seeking. Higher stakes may lead investors to make decisions that conform more closely to their preferences but deviate further from normative benchmarks. In other words, as simulated accounts are less important than real accounts, simulated account trades may be more random than real account trades.

The second mechanism through which stakes may strengthen biases is by strengthening the features of nontraditional preferences.⁶ Cognitive dissonance and self-image seem particularly relevant (Chang, Solomon, and Westerfield, 2016; Heimer, 2016).⁷ Investors may find that admitting to a mistake regarding a real account is much more embarrassing and wounding to one's self-image than admitting to a mistake with a simulated account. Higher cognitive dissonance provides a natural explanation for the real–simulated difference in the disposition effect (Chang, Solomon, and Westerfield, 2016). It may also explain other biases. When investors really want to do well, feel smart, and show off, they may trade more to make that happen. They may also hold a smaller number of stocks (especially volatile and positively skewed lottery stocks) where they believe gains are likely to be greater. In other words, investors might worry that diversifying their portfolios will equate to admitting they do not have good ideas and so want to hedge their bets. Cognitive dissonance may also explain the non-presence of a significant real–simulated difference in local bias. Such discomfort from cognitive dissonance largely involves an investor's self-image, and the decision whether or not to buy local stocks does not seem to relate strongly to self-image.

The first mechanism that links high stakes with strong biases predicts that higher stakes affect the translation of incorrect beliefs or nontraditional preferences into behaviors but do not predict changes in preferences per se. The second mechanism predicts changes in preferences. Section 2 (in particular Table 2) summarizes prominent interpretations of the studied biases and indicates how these mechanisms may play a role. Distinguishing between these mechanisms runs beyond the scope of this paper, but we provide some suggestive evidence that is consistent with both. For example,

⁶ For example, Rottenstreich and Hsee (2001) and Hsee and Rottenstreich (2004) show that probability weighting is more pronounced when lottery-like payoffs are more affect-rich. Compared with outcomes in simulated accounts, payoffs in real accounts may be more affect-rich, potentially explaining the findings pertaining to biases related to probability weighting.

⁷ Consistent with cognitive dissonance, Elia (2024) documents that hedge funds react to unrealized losses on their passive positions with activism.

both mechanisms predict that investors will exert greater effort on and pay more attention to their real accounts than their simulated accounts. Consistent with results reported in the experimental literature (Enke et al., 2021), we find compelling evidence that higher stakes generate greater effort/attention. Trading volume has been used as a proxy for attention (Barber and Odean, 2008). As discussed above, investors trade significantly more frequently in real accounts, suggesting that they pay greater attention to those accounts.

We take advantage of another data feature to examine whether attention strengthens biases. Our main tests focus on the period during which an investor carries both types of accounts. Most of our sample investors do not open the two types of accounts simultaneously. We conjecture that investors in our sample will pay more attention to their real (simulated) accounts when they carry only one type of account than when they carry both types of accounts. We predict that, if attention strengthens biases, investors in our sample will exhibit stronger biases in their real (simulated) accounts when they do not carry a second type of account than when they do carry a second type of account. Our empirical results support these conjectures.

The cognitive dissonance mechanism generates additional predictions that we can test. First, Chang, Solomon, and Westerfield (2016) document that, when investors trade actively managed mutual funds, they exhibit a reverse disposition effect instead of the disposition effect. They explain both the disposition effect on stocks and the reverse disposition effect on funds based on cognitive dissonance—whereby investors avoid realizing losses because they dislike admitting that past purchases were mistakes, but delegation reverses this effect by allowing the investors to blame fund managers instead. If stakes strengthen cognitive dissonance, we expect to find a stronger reverse disposition effect on equity mutual funds with high stakes. Consistently, in US brokerage data, we find a stronger reverse disposition effect on equity mutual funds with higher portfolio weights. Second, in our Snowball data, we find that, conditional on selling, investors are more likely to broadcast their selling of winners than their selling of losers. Such a winner–loser difference is more pronounced with real accounts than with simulated accounts, which is consistent with the fact that trading for real money is more relevant to self-image than simulated trading. These two findings are consistent with the notion that stakes strengthen the effects of cognitive dissonance and self-image

and support the conjecture that cognitive dissonance and self-image may explain the stakes effects. It is hard to explain these findings through other mechanisms.

A caveat is that we do not know, with certainty, why investors maintain simulated accounts. In an ideal experiment we would like investors to hold the two types of account for the same purposes. Snowball investors may, however, also use simulated accounts for purposes other than performance maximization, such as using them as “watch lists” for exploring or trading for fun. How likely is it that the real–simulated differences are caused by distinct purposes? First, we emphasize that our findings that biases are stronger in real accounts than in simulated accounts operate on many biases, which correspond to a variety of underlying theories, making it more likely that this is actually a direct effect of stakes and not of something else. For example, if investors use simulated accounts as watch lists to explore or for fun, we should expect that they take higher risks in simulated accounts than in real accounts, but the opposite is true in the data. Second, the existence of a real account may make it more likely that investors use their simulated accounts for purposes other than performance maximization. So, trading in a simulated account before one opens a real account is more likely to indicate what investors are thinking about doing with real money. Comparing trading in real accounts with simulated trading before an investor opens a real account yields similar results. Third, many investors offer short descriptions of the purpose of a given account. We document a similar result based on the subsample of investors who explicitly stated performance maximization as the purpose (for additional discussion, see Section 3.1 and Section 5.1.2). Overall, the results strongly point to a stakes-based interpretation. Although we cannot rule out all other possible interpretations, the results challenge the plausibility of such alternatives.

Our research contributes, in the first instance, to the literature on how stakes affect biases. To the best of our knowledge, we are the first to document a negative effect of stakes on real-world financial investment tasks. Even in the broader economics literature our study is novel in several respects. The idea that lab experiments should be dismissed because of their small economic incentives is a frequently cited claim but one that has not been widely tested. Very few studies use within-subject designs that can rule out population differences of this sort. And very few studies examine a range of mistakes or examine them in high-stakes environments. In the next section, we discuss the related literature in greater detail.

The remainder of the paper is organized as follows. Section 2 provides an overview of related research. We also discuss the mechanisms that might underlie our findings. In Section 3 we describe our data and methodology. In Section 4 we define our measures of investor biases and report summary statistics for those measures. Section 5 presents the results. In Section 6 we report results based on the two brokerage account datasets. We conclude in Section 7.

2. Overview of related research

We review the related literature on stakes effects and discuss differences between ours and related studies as well as plausible mechanisms that could explain our findings, including discussions of the roles stakes might play in affecting each bias. We also explain how our findings shed light on the mechanisms that underlie investor biases. To conclude, we discuss the external validity of our findings.

2.1 Most studies on stakes are lab studies

The great bulk of the stakes literature consists of experimental studies conducted in labs.⁸ As a result of budget considerations, the standard method for studying the effects of stakes on behaviors involves using payments that amount to a couple of hours of an individual's wages to incentivize experimental subjects—typically students—to commit time and effort to participating. In some studies, researchers can pay subjects up to several times their normal monthly incomes, typically when experimenting in low-income countries.⁹ Despite decades of research on the stakes effect, empirical evidence that compares biases under no or small stakes and those under very large stakes is scant (Enke et al., 2021). Our study provides novel evidence based on a non-experimental setting in which we compare the effects of no monetary stakes with the effects of high monetary stakes. We

⁸ See Camerer and Hogarth (1999) and Enke et al. (2021) for more comprehensive reviews of this literature.

⁹ Most of these studies on high stakes examine subjects' behaviors in strategic games such as the ultimatum game or coordination games. Slonim and Roth (1998), Cameron (1999), and Andersen et al. (2011) study the ultimatum game. Parravano and Poulsen (2015) study coordination games. Rapoport et al. (2003) study a three-person centipede game. Cooper et al. (1999) study gaming against managers. Binswanger (1980), Kachelmeier and Shehata (1992), and Holt and Laury (2002) study risk preferences. Kachelmeier and Shehata (1992) conducted a study in China where they can offer very large monetary incentives relative to the cost of living for most subjects. They report that the revealed risk preferences vary with the magnitudes of monetary incentives. Even with extremely high monetary incentives (when subjects earned three times their normal monthly revenues in the course of a two-hour experiment), subjects exhibit probability weighting.

study a standard set of prominent behavioral biases that financial market investors typically exhibit. The great bulk of this literature examines behaviors different from the behavior observed in our study.¹⁰

Field studies on stakes are almost always based on cross-individual comparisons. Chetty et al. (2014) show that individuals in Denmark are inattentive with respect to their pension contributions regardless of the economic stakes. Gathergood et al. (2019) find evidence that individuals' credit card debt repayments are not allocated to cards that charge higher interest rates, and the degree of misallocation is invariant with the economic stakes. Cross-individual comparisons are affected by omitted variables such as financial sophistication, wealth, and IQ. To the best of our knowledge, Andersen, Mukherjee, and Nielsen (2022) have conducted the only other field study that examines the causal effects of stakes. They exploit exogenous variation in stakes resulting from unexpected inheritances following sudden parental death and report economically small effects on biases, with most bias reductions passive (i.e., from an inheritance *per se*). For example, inheriting assets from parents mechanically reduces a beneficiary's underdiversification if the inherited assets are weakly correlated with the beneficiary's portfolio. In their setting, it is also difficult to differentiate between stakes effects and wealth effects, while wealth effects have been shown to affect stock market participation in the same setting (Andersen and Nielsen, 2011).

2.2 The effects of stakes depend on the nature of the tasks involved

In contrast to conventional wisdom, according to which higher stakes should reduce biases, existing studies show that the effects of stakes on behavior and performance depend on the nature of the tasks involved. Table 1 provides a summary of this literature. We classify existing studies into five groups based on stakes effects—the first three involve mainly beliefs and judgments, while the

¹⁰ There is also a body of literature documenting that some biases survive in the high-stakes real world. Individuals exhibit various biases on game shows (Metrick, 1995; Berk et al., 1996; Levitt, 2004; Post et al., 2008; Belot et al., 2010; Van den Assem et al., 2012; Jetter and Walker, 2017), in art auctions (Beggs and Graddy, 2009; Graddy et al., 2014), in professional golf competition (Pope and Schweitzer, 2011), and among asylum judges, loan officers, and baseball umpires (Chen et al., 2016). The behavioral finance field has documented many biases (Barberis and Thaler, 2003; Campbell, 2006; Barber and Odean, 2013; Beshears et al., 2018; Malmendier, 2018) that we study. Our contribution is not to document the existence of biases in high-stakes financial markets but to document that, in real-world financial markets, higher stakes can strengthen biases, while the studies cited in this note do not compare what occurs in high- and low-stakes situations.

last two involve mainly preferences. We further classify three groups into subgroups to reflect their relatively distinct mechanisms.

First, the first group of studies indicates that higher stakes improve performance. Such findings concentrate on tasks where greater effort can improve performance (Camerer and Hogarth, 1999). Such tasks include memory or recall tasks as well as clerical tasks (e.g., coding words or building things). There is also some evidence that stakes improve academic performance (Levitt et al., 2016). In a recent study, Enke et al. (2021) find that higher stakes also help subjects taking the Cognitive Reflection Test, which is a task designed to measure a person’s tendency to override an impulsive incorrect response and engage in further reflection to find a correct answer.¹¹ Second, stakes do not change average behavior substantially with respect to tasks involving high cognitive effort costs, such as tasks testing base rate neglect and anchoring (Camerer and Hogarth, 1999; Enke et al., 2021).

Third, stakes can hurt performance. Such phenomena were typically found with tasks that involve creativity or tasks in which simple intuition or habit provides optimal answers and more extensive effort in thinking about options makes things worse (Arkes, Dawes, and Christensen, 1986; Ashton, 1990; Hogarth et al., 1991; Ariely et al., 2009). These findings suggest that individuals may “choke under pressure” (Baumeister, 1984; Ariely et al., 2009). “Choking under pressure” is most likely to happen when the pressure a situation imposes on an individual overwhelms his or her capacity to cope with it. Such pressure causes individuals, involuntarily, to think consciously about the task and worry, reducing the mental capacity to process task-relevant information (Baumeister and Showers, 1986). “Choking under pressure” can explain why stakes hurt performance, but the mechanism does not seem particularly relevant in our financial context. For instance, “choking under pressure” is typically associated with stress and anxiety, in sharp contrast with the finding that investors are typically overconfident (Odean, 1998).¹²

Another mechanism through which stakes might hurt performance occurs when subjects form incorrect mental models (Dawes, 1979; Dawes, Faust, and Meehl, 1989). Raising stakes stimulates

¹¹ Enke et al. (2021, p33-34) write that, “. . . reducing reliance on ‘system 1’ is not enough to overcome biases because these are often limitations of the more deliberative ‘system 2.’”

¹² Additionally, “choking under pressure” is typically associated with unpracticed tasks (Beilock et al., 2004), while a typical investor in our sample trades frequently.

greater attention and effort, facilitating the translation of incorrect beliefs into behaviors. A recent experiment conducted by Hartzmark, Hirshman, and Imas (2021) documents that individuals react in more extreme ways to information about a good they own than to the same information about a non-owned good, consistent with the idea that investors harbor incorrect beliefs, with the result that attention actually hurts those investors. Giglio et al. (2021) document direct evidence that attention strengthens the sensitivity of portfolios to beliefs. Although these studies focus on belief-updating models, the same arguments can apply to cases involving nontraditional preferences, including preferences that have been proposed to explain many of the studied biases (such as prospect theory, realization utility, cognitive dissonance, and sensation-seeking). Higher stakes may lead investors to make decisions that conform more closely to their beliefs or preferences but deviate further from normative benchmarks.

Fourth, high stakes affect behaviors involving nonpecuniary considerations. High stakes seem to reduce the weight of nonpecuniary factors such as moral costs or reputational concerns and shift subjects' behaviors in strategic games toward the predictions of rational-choice models (Smith and Walker, 1993; Camerer and Hogarth, 1999; Levitt and List, 2007; Andersen et al., 2011). Andersen et al. (2011) find that, in the ultimatum game, sufficiently high stakes lead responder behaviors to converge almost perfectly on full acceptance of low offers. In some cases, extrinsic monetary incentives crowd out intrinsic incentives such as volunteering and other pro-social behaviors (Gneezy and Rustichini, 2000; Gneezy et al., 2011; Bowles and Polania-Reyes, 2012).

Fifth, several studies examine the effects of stakes on risk preferences. Tversky and Kahneman (1992) find no significant effect of stakes on risk preferences. More recent studies, however, find a significant stakes effect: when the stakes are high, individuals are risk-averse in gains and risk-seeking in losses, but risk preferences reverse when the stakes are low, with individuals exhibiting risk-seeking in gains and risk-aversion in losses (Fehr-Duda et al., 2010; Scholten and Read, 2014; Khaw, Li, and Woodford, 2022).¹³ Such stakes effects do not seem to explain our findings. For example, an increase in stakes is expected to be associated with a decrease in demand for lottery-type stocks, which is the opposite of what we find.

¹³ Such a pattern in risk preferences had been conjectured as early as in Markowitz (1952).

The “risk as feeling” literature has also identified a relevant mechanism. Higher emotion strengthens the elements of prospect theory (Rottenstreich and Hsee, 2001; Hsee and Rottenstreich, 2004). Faro and Rottenstreich (2006) find that individuals predict that others hold less pronounced prospect-theory preferences than actually prevail, and Andersson et al. (2016) find that deciding for others reduces loss aversion; both of these findings are consistent with the notion that deciding for others or predicting what others will do is less emotional than deciding for oneself or predicting one’s own behavior. It is natural to think that trading for real money is more emotional than trading in simulated accounts and higher emotion can potentially lead to a more pronounced value function and a more pronounced probability-weighting function. A more pronounced value function and a more pronounced probability-weighting function can, in turn, lead to a stronger disposition effect (Shefrin and Statman, 1985; Odean, 1998; Barberis and Xiong, 2009) and greater demand for lottery stocks (Barberis and Huang, 2008). A stronger lottery preference may also explain underdiversification because an investor may need to sacrifice diversification to gain portfolio skewness (Barberis and Huang, 2008).

2.3 Mechanisms that are most relevant to our findings

Table 2 summarizes prominent interpretations of the studied biases and the roles that stakes may play. Perhaps the most promising mechanisms that could explain our findings are the incorrect mental model mechanism (including nontraditional preferences) and the stakes-strengthening nontraditional preferences mechanism. These two mechanisms are not mutually exclusive, and both likely play a role.

The incorrect mental model mechanism can potentially explain the results for all the biases we study, as it applies to both belief-based and preference-based biases. As discussed above, higher stakes may strengthen the features of prospect theory and help explain prospect theory–related biases. Higher stakes may also strengthen the effects of cognitive dissonance and self-image (Chang, Solomon, and Westerfield, 2016; Heimer, 2016) and help explain the findings for the disposition effect, underdiversification, turnover, and lottery preferences as well as the non-result for local bias. The findings for the reverse disposition effect and investor postings can be explained naturally by

reference to cognitive dissonance and self-image but are hard to explain by reference to other mechanisms.

2.4 Investor biases

Our study also contributes to the literature that studies mechanisms that underlie investor biases. Any explanation of a bias should also be able to explain why the bias worsens at higher stakes or at least be consistent with this prediction. Cronqvist and Siegel (2014) document that genetic differences explain a large portion of cross-investor variations in biases.¹⁴ Grinblatt, Keloharju, and Linnainmaa (2011, 2012) show that underdiversification and the disposition effect are stronger in individuals with lower IQs. Barberis (2013) argues that prospect theory can explain several investor biases. Gabaix (2014, 2019) provides a unifying framework for considering inattention as a common source of several biases, and Birru et al. (2024) provide evidence that is consistent with this framework, which assumes that individuals carry the correct mental models but cognitive costs lead them to form noisy expectations. Such a bounded rationality framework predicts that higher stakes will induce greater effort and hence reduce biases. Our findings and those of several recent studies (Enke et al., 2021; Hartzmark, Hirshman, and Imas, 2021) suggest that many individuals may not carry the correct mental models, and that poor decisions are driven in part by deeply rooted biases that strong motivations, at least alone, cannot improve.¹⁵ Our findings also shed light on the existence—and persistence—of so many biases and asset-pricing anomalies in high-stakes real-world financial markets.

Our paper also addresses investor protection policies. If investor biases are deeply rooted, policies that make trading easier may hurt investor welfare, as evidenced when investors switched from phone-based to online trading in the 1990s (Baber and Odean, 2002) and more recently to smartphone-based trading (Kalda et al., 2023). In this spirit, our finding is also related to those of Heimer and Simsek (2019) and Heimer and Imas (2022), who document that leverage constraints can

¹⁴ Cronqvist and Siegel (2014) also review the literature on the association of specific genes with biases.

¹⁵ Evolutionary models of biases provide insights into this phenomenon. Proponents of these models argue that behaviors that represent investment biases today might have been advantageous in ancient times (Rayo and Becker, 2007).

discourage retail investors from trading excessively, weaken the disposition effect, and improve welfare.

2.5 External validity

Finally, we stress that we do not expect our stakes-strengthen-biases results to extend to all other biases or all other settings. As discussed above, stakes have effects that vary with the nature of specific tasks. When discussing the generalizability issue, it is useful to emphasize one important feature of financial markets. Stock returns are characterized by a low information-to-noise ratio. Such a feature may prevent investors from learning. Accurate learning takes place only when individuals receive timely and organized feedback (Thaler, 1986). Competition does not seem to work effectively at eliminating biased investors from financial markets (De Long et al., 1990). In addition, in financial markets, nonpecuniary factors may not exist or are unimportant factors in investor trading. As discussed above, evidence shows that high stakes tend to reduce the consideration of nonpecuniary factors. Last, our data are drawn from an investor social network and, as a result, our investors' trades can be observed by their peers. Although our within-subject estimates remain valid because both types of accounts are observable, we note that stakes effects may be a function of context. The biases in our real accounts are similar to what have been documented using other account-level datasets, and we find similar results for investors who have no followers, mitigating this concern. Our analysis based on the two additional account-level datasets confirms the finding that stakes strengthen biases, providing external validity to the real–simulated account comparison analysis.

3. Data and methodology

Our primary data source is Snowball Finance (Xueqiu.com; Snowball hereafter), supplemented with stock price, return, and volume data from the China Stock Market & Accounting Research Database (CSMAR).¹⁶ In this section, we discuss background information pertaining to Snowball and the data.

¹⁶ Sui and Wang (2025) also use Snowball data. They focus on real accounts and study self-enhancing transmission bias.

3.1 Snowball Finance

Snowball is one of the most popular social networks among Chinese investors. Founded in 2010, it has 12 million active users. Although the users range from small retail investors to professionals, including sell-side analysts and fund managers, Snowball is well known to be overpopulated with relatively sophisticated retail investors. On Snowball, investors can access real-time price and volume information on financial securities. Like Twitter, Snowball allows its users to share information, post their opinions on investment and financial issues, and interact with other users.

Uniquely, Snowball allows users to carry two types of investment accounts: real and simulated. Users can register at Snowball and obtain one or more simulated accounts. They can also link their brokerage accounts with Snowball and reveal their transactions on the website. Third-party brokerage companies manage the brokerage accounts, and Snowball does not profit from trades made by these investors.¹⁷ Around half a million users hold stock accounts with Snowball, either real or simulated.

Users may link their brokerage accounts for several reasons, and some benefits and website functions are available only if they do. For example, only linked investors can observe other linked investors' trading records. Through collaboration with brokerage firms, Snowball helps linked users participate in IPO subscriptions and enhance the returns on their cash balances (i.e., by participating in reverse repurchase agreements or money market mutual funds). Investors may also enjoy the experience of interacting with others and feeling influential.

Although Snowball's simulated accounts do not involve real money, representatives of the platform informed us that most investors trade in simulated accounts to maximize performance. Many investors post short descriptions of the purposes of their simulated accounts. Our analysis of these purpose descriptions confirms what Snowball representatives said. This is not surprising, given Snowball's design. To incentivize investors to maximize performance with simulated accounts, Snowball maintains three lists of top-ten accounts. The three lists are based on performance in the preceding day, week, and year. There are similar lists for real accounts. The lists of best performers are updated in real time and featured prominently on the Snowball homepage. Moreover, Snowball

¹⁷ Most of the brokerage accounts are held at Ping'an Securities, one of the largest securities firms in China.

records are verifiable and many users report the performance they achieve with both types of accounts on their résumés.

Another possible utility that simulated accounts might provide would involve using them as “watch lists.” Almost all brokerage trading platforms in China (including brokerages that manage real Snowball accounts) offer investors a dedicated “watch list” function. Therefore, Snowball users do not have to use simulated accounts as “watch lists.” Consistently, we find that, across all purchases in the real accounts, only 16.8% of the purchased stocks were already held in simulated accounts at the time of purchase.

Snowball follows real stock market rules as closely as possible to manage trading in simulated accounts. There are, however, several differences. First, orders from simulated accounts are matched with the prevailing exchange order book, but simulated trades will not change the order book or have a price impact.¹⁸ Second, Snowball does not consider brokerage commissions (about 0.25% of the value of a transaction) or stamp-duty taxes (0.1% of transaction value, for sales only) in calculating the values of positions in the simulated accounts. Third, cash dividends are automatically reinvested in those dividend-paying stocks.

We do not expect these differences to have important effects on our results. The sample investors are retail investors. Treating them as price-takers (i.e., with no impact on an order book) is approximately correct. Given that Snowball does not consider commissions or taxes with simulated accounts, we expect investors to trade more actively in those accounts. As we show later, however, investors trade significantly more frequently in real accounts.

3.2 Snowball data

The data we collected from Snowball are very similar to those used by Odean (1998) and Barber and Odean (2000). We created a user file and a trade file. We also collected a file with all the posts. The first real account was opened on June 24, 2016 and the first simulated account was opened on October 22, 2014. Our sample period starts on June 24, 2016 and ends in May 2019, when we began collecting data.

¹⁸ Both the Shanghai Stock Exchange and the Shenzhen Stock Exchange are pure limit-order markets.

At the user level, we collected information on user identifications, the list of accounts a user holds (including closed accounts), a real/simulated flag for each account, an active/closed flag for each account (measured at the time we collected data), an account's initiation date, an account's close date (if applicable), and user locations if available. About half of the users in our main sample reported their locations, either at the province level or the prefecture-city level.

We also created a trade file. The trade file contains such information as investor identification, account identification, trade date, trade time, asset symbol, type of record (dividend payment, buy/sell, IPO subscription, etc.), and trade price.¹⁹ The data do not include information indicating the number of shares for each asset or the total value of an account. Instead, the data include the value of one asset as a fraction of the total portfolio value. For each trade, we collect the fractions before and after the trade. We infer the direction of a trade by comparing the change in fractions. For example, if the fraction of a stock drops from 50% to 20% after a trade, we infer that this is a sale. Information indicating changes in fractions also enables us to calculate portfolio turnover. The sum of a change in fractions within a given period gives us a measure of portfolio turnover.

Once the link between an investor's brokerage account and Snowball is set up, all subsequent real-account trades will be disclosed on Snowball. Trades that occur before the link are not disclosed, which avoids selective disclosure based on ex-post performance. All trades from the simulated accounts are disclosed.

The data are subject to several limitations. First, users can close their simulated accounts or real accounts at any time. Trading records and daily portfolio performance data for closed simulated accounts are kept on the website but are removed for closed real accounts. For closed real accounts, the website keeps account initiation dates, account close dates, and portfolio performance as of account close dates. The fact that the trading histories of closed real accounts are unavailable to us raises a sample-selection concern. We argue that this sample-selection concern may not be particularly problematic. First, Sui and Wang (2025) find that real account closings do not seem to

¹⁹ We use raw trade prices before fees or commissions. The two Chinese stock exchanges are both pure limit- order book markets. Chinese brokerage companies collect client orders and pass them to the exchanges. Fees and commissions are collected separately, not through spreads. We selected 200 transactions randomly and compared them with prices from intraday trade-by-trade data provided by RiceQuant. In almost all cases, the prices are exactly the same or within a one-cent difference.

correlate with performance. Second, as documented later, investors achieve *worse* performance and exhibit *stronger* biases in their real accounts than in their simulated accounts. If an investor is more likely to close a real account when performance falls or when biases are stronger, we would expect the sample selection process to lead to underestimation of the stakes effects.

Second, Snowball sets limits on the total number of trades available on the website. If the number of trades from a real account is above 200, only the most recent 200 trades will be visible. The limit is 1,000 for simulated accounts. The majority of accounts do not reach these limits, and therefore we have data on their entire trading histories for most of the accounts in our sample. Truncation is more likely to affect users who trade very frequently. Our data overweight their trading in the period immediately before May 2019, when we collected our data. We evaluate how this truncation affects our results later in the paper.

3.3 Summary statistics

We focus on investors who hold both real and simulated accounts. In our main analyses, for each investor who holds accounts of both types we restrict our analysis to the period when both types of accounts are active. In some tests, we examine whether the presence of the other account affects the biases. To collect sufficiently voluminous data to measure investor behaviors, we require that the overlapping period lasts for at least one year.

We start with 13,993 investors with real accounts. Of these investors, 8,732 also hold simulated accounts. To be included in our sample, an investor has to trade stocks listed on the two domestic exchanges in both types of accounts. As a result, 398 investors are excluded. Lastly, we exclude 3,921 investors whose real and simulated accounts overlap for less than one year. Our final sample includes 4,413 investors. Virtually every investor holds one real account, but many hold multiple simulated accounts.²⁰ In total, the 4,413 investors hold 20,603 simulated accounts. Among the final sample, 1,081 investors hold only one simulated account. Our results are similar if we focus on these 1,081 investors.

²⁰ One common reason for holding multiple simulated accounts is to trade different types of stocks in separate accounts. For example, one can have an account for value stocks and another for momentum stocks.

In Table 3, we report summary statistics for the sample investors. The average length of the overlapping period is 668 days. The average number of trades in the real accounts is 99.66. For investors with multiple simulated accounts, we first calculate the number of trades for each simulated account and then average it across all the simulated accounts an investor holds. We obtain an average number of trades in a simulated account of 39.77.

We also report how often investors hold the same stocks in both types of accounts. Overlapped holdings are the number of stock-days when an investor holds a stock in both types of accounts on the same day. Real holdings (simulated holdings) are the number of stock-days in real (simulated) accounts. On average, the overlapped holdings are 35.6% of all real account holdings, indicating that many investors carry out a similar thought process when trading in both types of accounts. The ratio is much smaller as a fraction of simulated account holdings because more stocks are held in the simulated accounts. It is unsurprising to see overlapped holdings as retail investors have limited attention. We also observe sizeable cross-sectional variation in the ratio of overlapped holdings. In Section 5.1.2, we report that our main results are robust in all subsamples split by this ratio.

Although Snowball does not reveal account values, our conservative estimates suggest that the average real account value is worth four to six times the annual disposable income per capita in China. We estimate the value of each real account by taking advantage of the exchange trading rule, according to which a stock's minimum trading unit is 100 shares. First, we estimate the value of each trade by assuming that the trade size is 100 shares. This is the lowest possible trade value. Second, Snowball reports the size of each trade as a fraction of a trader's total portfolio value. We divide the trade value by that fraction, obtaining an estimate of the lower bound of the portfolio value at the time of the trade. Third, we calculate the highest portfolio value across all the trades executed by a given investor and use this figure as our final measure of portfolio value. The estimated mean (median) portfolio value is 244,000 RMB (106,000 RMB) or approximately 35,030 US dollars (15,218 US dollars). The mean (median) annual disposable income per capita in China in 2019 (the last year of our sample) was 30,733 RMB (26,523 RMB) or approximately 4,412 US dollars (3,808 US dollars).

4. Investor behaviors and methodology

In this section, we introduce the investor behaviors that we examine in this paper. All the behaviors we examine have been widely studied in the finance literature (Barberis and Thaler, 2003; Barber and Odean, 2013). For some behaviors, there may be alternative measures. We discuss these alternative measures in the Internet Appendix, where we report the results of robustness tests conducted with these measures.

We calculate each investor behavior measure at the individual level and separately for real accounts and simulated accounts. As noted above, virtually all the investors in our sample hold just one real account, but many hold multiple simulated accounts. We estimate the disposition effect by pooling data from all the simulated accounts together. We first calculate other bias measures for each simulated account and then take the average across all of an investor's simulated accounts. For diversification analysis, in addition to the average-across-accounts approach we also measure diversification by treating all the simulated accounts as one and report this result in the Internet Appendix. Our results are qualitatively similar if we focus on investors who hold just one simulated account. To mitigate the effects of extreme values, we winsorize all the measures at the 1% and the 99% levels separately for real accounts and simulated accounts.

4.1 Investor behaviors

4.1.1 The disposition effect

The disposition effect—the tendency to sell stocks that earn capital gains rather than stocks that incur capital losses—is perhaps the most robust empirical finding on investor trading (Shefrin and Statman, 1985; Odean, 1998).

To examine whether investors in our sample are likely to sell winner stocks and hold loser stocks, we first use the transaction dataset to construct a holding sample that contains observations for all investor-stock-day pairs (Ben-David and Hirshleifer, 2012). For example, if an investor bought Stock A on January 2, 2018, and held it until January 31, 2018, there will be 22 observations (22 business days). Purchases and sales are aggregated on a daily basis for each account-stock. We flag the days when a position is opened and when shares are sold (including partial sales). We use the

weighted average purchase price and take stock splits and dividends into account to calculate the holding-period returns. We exclude positions for which we do not have the initial purchase price, mainly because those positions were established before a Snowball link was made.

We estimate the disposition effect using the following model,

$$Sell_{i,j,t} = \alpha + \beta_i Gain_{i,j,t-1} + \varepsilon_{i,j,t}, \quad (1)$$

where i, j , and t denote investor i , position j , and day t , respectively. $Sell_{i,j,t}$ is a dummy variable that equals 1 if investor i sells position j (partially or fully) on day t and 0 otherwise. $Gain_{i,j,t-1}$ is a dummy variable that equals 1 if investor i experiences a gain in stock j at the end of day $t-1$ and 0 otherwise. We estimate Equation (1) using a linear probability model. The coefficient, β_i , is our measure of the disposition effect for investor i . To facilitate the presentation of the reported results, we multiply β_i by 100.

An alternative specification involves focusing on investor-days when an investor sells at least one stock and has at least two stocks available for sale (Odean, 1998). The individual-level disposition effect estimates may be noisy. We could, alternatively, estimate the disposition effect by pooling all real accounts or all simulated accounts together. We can also estimate Equation (1) using a hazard model instead of a linear probability model. We find qualitatively similar results with all these methods and report them in the Internet Appendix.

4.1.2 Lottery preference

We define lottery stocks following Kumar (2009). Specifically, we consider three stock characteristics to identify stocks that might be perceived as lotteries: idiosyncratic volatility, idiosyncratic skewness, and stock price. At the end of month t , we compute both idiosyncratic volatility and idiosyncratic skewness using daily data from $t-6$ through $t-1$. The idiosyncratic volatility measure is the variance of the residuals obtained by fitting a one-factor model to the daily stock returns, where the factor is market excess returns. We measure idiosyncratic skewness following Harvey and Siddique (2000). Specifically, idiosyncratic skewness is a scaled measure of the third moment of the residuals obtained by fitting a two-factor model to daily stock returns, where the two factors are excess market returns and squared excess market returns.

We consider all listed domestic Chinese stocks and define stocks in the lowest 50th stock price percentile, the highest 50th idiosyncratic volatility percentile, and the highest 50th idiosyncratic skewness percentile as lottery-type stocks (Kumar, 2009).

At the individual investor level, we calculate the strength of lottery preferences separately for real accounts and simulated accounts, using the percentage of lottery stocks an investor buys over the sample period. We weigh each purchase with a purchase's dollar amount as a fraction of the total portfolio value at the time of purchase. For example, if a purchase represents 20% of the total portfolio value at the time of purchase and another purchase represents 40% of the total portfolio value at the time of purchase, the second purchase is assigned a weight that is twice that of the first purchase. We obtain qualitatively similar results if we weigh separate purchases equally or measure lottery preferences based on holdings instead of purchases.

We report the results based on the three characteristics (price per share, idiosyncratic volatility, and idiosyncratic skewness) in the Internet Appendix.

4.1.3 Extrapolation

Investors tend to buy stocks that have abnormally high past performance (Barber and Odean, 2013). We expect investors harboring stronger extrapolation bias to buy stocks that have earned higher recent past returns. For each stock an investor bought on day t , we calculate the excess return on this stock, defined as the return on the stock net of the market return over the preceding 126 trading days. We aggregate these excess returns (in percentages) across all of an investor's purchases separately for real and simulated accounts. As we do for the lottery-preference measure, here we weigh each purchase by its dollar amount as a fraction of the total portfolio value at the time of purchase. We use average past excess returns as the proxy for investors' tendency to follow extrapolation strategies. Our results, which we report in the Internet Appendix, are not sensitive to window choices.

4.1.4 Underdiversification

Our measure of diversification is the number of stocks in an investor's portfolio. We count the number of stocks in a portfolio every day and use the time-series average as our measure of

diversification. For all other bias measures, a higher value is associated with a stronger bias. To be consistent, and for ease of exposition, we use the negative value of the natural logarithm of the average number of stocks as our measure of underdiversification.

4.1.5 Local bias

It has been documented that investors prefer holding stocks in firms that are located geographically near to them (Ivkovic and Weisbenner, 2005; Massa and Simonov, 2006).

To test whether investors in our sample tend to hold local stocks, we collect the headquarters locations of all the listed firms and the residence cities of the investors.²¹ We obtain firm headquarters locations from WIND. We then identify the latitude and longitude of each location by relying on the Baidu map Application Programming Interface and calculate the distance (in kilometers) between the headquarters cities and investors' residence cities using the haversine formula. The haversine formula determines the great-circle distance between two points on a sphere, given their longitudes and latitudes.

Our primary measure of local bias is the benchmark-adjusted average distance (in kilometers) between an investor's location and the headquarters of the stocks the investor bought over our sample period. We first calculate the volume-weighted average distance between an investor and his/her purchased stocks. The benchmark is the market capitalization-weighted average distance between this investor's location and the headquarters locations of all the listed stocks. Investors for whom the average distance is shorter than the benchmark are more likely to exhibit local biases. Our main local bias measure is calculated as $\log(1 + \text{benchmark distance}) - \log(1 + \text{average distance})$. We add 1 to the average distance because there are cases where investors buy only stocks located in their cities of residence and therefore exhibit an average distance of 0. Our results are also similar if we measure local bias using the excess weight by which an investor buys stocks in his/her own province or prefecture city.

²¹ About 50% of our sample investors report their locations. Of the investors who report their locations, about 80% report at the prefecture-city level and the other 20% report at the province level. We drop the latter from the analysis when calculating the benchmark-adjusted distance and when calculating the excess weight of stocks in an investor's city of residence.

4.1.6 Turnover

Barber and Odean (2000) find that investors trade too frequently. We investigate the frequency of trading for the investors in our sample using portfolio turnover.

We calculate turnover at the account level through the following steps. First, we define the date for the first and last observations in one portfolio within the overlapping period as the start and end dates. We obtain the duration of one portfolio as the total number of days between the end date and the start date. Notice that we measure duration in calendar days rather than trading days. We aggregate the corresponding fractions across all trades within the duration of one portfolio and divide the sum by two. Finally, we obtain the daily account turnover by dividing the aggregated fraction by the total number of duration days. We report the annualized turnover rate by multiplying daily turnover by 365.

4.1.7 Investor performance

For each purchase we calculate the stock's buy-and-hold returns (in percentages) over the following two months. We adjust this figure by size-decile portfolio returns. As we do for lottery preferences, extrapolation, and local bias, here we weigh each purchase with the purchase's dollar amount as a fraction of the total portfolio value at the time of purchase. Our results are similar if we measure performance over the following one, three, or six months. Size-decile portfolios are formed based on market capitalization at the end of the previous month and rebalanced monthly.²² Our performance measure is not affected by the difference in the treatments of commissions and taxes between the real and simulated accounts.

²² Snowball reports portfolio performance on its website. As stated on the site, though, its calculation "may be affected by special cases." Our calculation shows that there are an unusually high number of days with extreme returns. For example, about 14% of real portfolios experience a daily return that is higher than 22.22% or lower than -22.2%. With the daily $\pm 10\%$ price-limit rule, 22.22% and -22.2% are the maximum and minimum possible daily portfolio returns. Our further analysis suggests that, in some cases, Snowball does not properly observe cash flows into or out of an account. Our private communications with Snowball confirm this conjecture. Therefore, we decided to calculate portfolio returns by ourselves instead of using the reported returns.

4.1.8 Discussions

Most investor behavior literature regards the abovementioned behaviors as suboptimal. The theoretical literature has proposed various mechanisms to explain these behaviors based on irrational beliefs, non-traditional preferences, or bounded rationality. The empirical evidence shows predominately that these behaviors are associated with inferior investor performance.²³ For example, Barber, Lee, Liu, and Odean (2009) show that, in aggregate, the portfolios of individual investors in Taiwan suffer an annual performance loss equivalent to 2.2% of Taiwan's gross domestic product, and at least two-thirds of the losses (commissions and transaction taxes paid) are attributable to heavy trading.

Evidence from mainland Chinese data also points in the same direction. For the disposition effect, studies have argued that it could be motivated by a desire to rebalance portfolios or to avoid the higher trading costs associated with low-priced stocks. These rational interpretations have been dismissed both in the US data and in mainland Chinese data (Odean, 1998; An et al., 2024). Lottery stocks underperform other stocks both in the US (Kumar, 2009) and in mainland China (Jiang et al., 2022) and therefore a lottery preference is costly. Investors can exhibit extrapolation (i.e., buy past winners) if past winners continue to perform better (i.e., return momentum). The literature reports no momentum effect or a reverse-momentum effect in China (Liu, Stambaugh, and Yuan, 2019; Jiang et al., 2022). Investors can exhibit local bias if they are well informed about local stocks, a finding that is not supported by the data either in the US (Seasholes and Zhu, 2010) or in mainland China (Seasholes, Tai, and Yang, 2010). Not surprisingly, as we show later, all these biases are negatively correlated with investor performance.

4.2 Summary statistics and correlation coefficients of biases

In Table 4, we report the summary statistics for the biases. For each behavioral bias measure, we report, separately for the real and simulated accounts, the first quartile, the median, the third quartile, and the number of investors for whom we can construct the measure as well as the means and standard deviations. It is evident that, in both simulated and real accounts, investors exhibit a

²³ See the references in the Introduction as well as Barberis and Thaler (2003) and Barber and Odean (2013) for reviews of both the theoretical and empirical contributions to this literature.

strong disposition effect, extrapolation, and local bias. They also hold a small number of stocks in both types of accounts. The average fraction of lottery stocks in the entire stock market is 6.09%, so our investors exhibit a strong lottery preference in real accounts but not in simulated accounts. Investors earn negative abnormal returns in both types of accounts.

The biases in real accounts are qualitatively similar to those documented in the literature: the disposition effect (Feng and Seasholes, 2005; Frydman and Wang, 2020; An et al., 2024), lottery preference (Liu et al., 2022), extrapolation (Liao, Peng, and Zhu, 2022), underdiversification (Feng and Seasholes, 2008; Frydman and Wang, 2020), local bias (Feng and Seasholes, 2008), turnover (Frydman and Wang, 2020; Liu et al., 2022), and performance (An et al., 2024). In cases where the estimates are comparable, the biases in the real accounts are also quantitatively similar: the disposition effect (Frydman and Wang, 2020), underdiversification (Feng and Seasholes, 2008; Frydman and Wang, 2020), turnover (Liu et al., 2022), and performance (An et al., 2024). Such a finding mitigates the concern that our online investors may not be representative of the relevant population of investors. This finding also mitigates the concern that the observability of the trades on the online investor social network may dramatically change investors' behaviors.

In Table 5, we report the pairwise correlation coefficients between behavioral biases in real accounts (Panel A) and in simulated accounts (Panel B). First, in general, the biases are positively correlated with each other and are negatively correlated with performance. Most of the correlation coefficients are statistically significant. One exception is local bias, whose correlations with other variables are weak and do not exhibit a clear pattern. Local bias also does not strongly correlate with performance. The general negative correlations between the biases and performance are consistent with the view that these investor behaviors involve costly mistakes. Second, the correlations are not very strong. They are all lower than 0.3, suggesting that these biases are distinct from each other.²⁴ Third, the correlation coefficients between biases/performance are higher in real accounts than in simulated accounts.

²⁴ The generally weak positive correlations between investor biases and weak negative correlations between investor biases and performance is consistent with findings reported in the literature. See Bailey, Kumar, and Ng (2011) for an example. Liu et al. (2022) report a similar correlation table but their investor biases are calculated based on survey data. Attempts to consolidate many biases into fewer dimensions have also been made in the experimental economics literature (see Dean and Ortaleva, 2019; Chapman et al., 2023; Stango and Zinman, 2023), but these studies typically examine biases other than those we examine.

5. Results based on Snowball data

5.1 Stakes effects: Differences between real and simulated accounts

5.1.1 Main results

Table 6 presents the study’s main results. We conduct the within-subject comparison at the level of biases between real accounts and simulated accounts. For each behavioral bias measure, we report the means of both real accounts and simulated accounts. We also report the means of investor-level differences, their t -statistics, and Wilcoxon p -values.

Investors exhibit stronger biases in their real accounts than in their simulated accounts, with the exception that they do not exhibit a difference in local bias between the two types of accounts. These differences are highly statistically significant, as indicated by the high t -statistics. The differences are also sizable. In real accounts, the probability that investors sell winners is, on a daily basis, 4.16% higher than the probability that they sell losers. In the simulated accounts, the difference is only 0.49%. In the real accounts 6.49% of purchases involve lottery stocks, while in the simulated accounts 4.26% of purchases involve those stocks. On average, investors hold 2.96 (i.e., $\exp(1.086)$) stocks in their real accounts and 4.36 (i.e., $\exp(1.473)$) in their simulated accounts. Annual turnover is 596% in real accounts and 84% in simulated accounts. In the two months after being purchased, the stocks investors hold in their real accounts underperform their size-decile portfolio by 0.656%, while the stocks they hold in their simulated accounts underperform that portfolio by 0.148%.

In Figure 1, we present the estimated kernel densities for behavioral biases in real and simulated accounts. Each plot uses a Gaussian kernel. The distributions of the biases reveal the contrast between real and simulated accounts. With respect to all the biases except local bias, the distribution of real accounts exhibits a clear rightward shift from the distribution of simulated accounts. The performance distribution of real accounts exhibits a clear leftward shift. These graphs make it clear that the differences between the two types of accounts are sizable and systematic, echoing the strong statistical power associated with Table 6.

Investors' trading decisions can deteriorate for two reasons: they conduct more bad trades or each trade is worse. Our results show that both contribute to stronger biases and worse performance in real accounts. The higher turnover observed in real accounts indicates that stakes increase the frequency of mistakes. Given that we measure several biases (lottery preference, extrapolation, and performance) on a per-trade basis, the results also show that the average trade quality is lower in real accounts.²⁵

5.1.2 Robustness tests for the main results

We report the results of several robustness tests for the baseline results.

A. *Using simulated accounts that explicitly identify performance maximization as the purpose*

As discussed in Section 3.1, most simulated accounts are created to maximize performance. In this section, we present a robustness test in which we keep only the subsample of simulated accounts where investors *explicitly* identify performance maximization as the purpose. Many investors have a short description of the purposes of their simulated accounts. We hired eight students to read through all the descriptions. For each account we assigned two students to read its purpose description independently. Our students were asked to minimize false positives when identifying accounts for performance-maximization purposes. Ultimately, we classify an account as designed for performance maximization only if both students identify this purpose as the rationale. Of 21,866 simulated accounts, 6,357 are classified as explicitly stating performance maximization as the purpose. Please note that most others include purpose statements that are either too short or otherwise uninformative.²⁶ In Internet Appendix Section A, we present the baseline analysis based on this subsample and document a result similar to that reported in Table 6.

²⁵ The two types of accounts differ dramatically in turnover. The disposition effect measure can be directly affected by turnover (Feng and Seasholes, 2005). We find that such a mechanical relationship does not drive our results because we find similar results using an alternative disposition effect measure that is free of such a mechanical relationship. See Appendix Section H for additional discussion and results. Many other biases are not directly affected by turnover. For example, the lottery preference and extrapolation measures are averaged across purchases and are not functions of sales or the frequency of purchases.

²⁶ In addition to performing manual classification, we adopt a keyword-based classification method. We compile a list of keywords from all the purpose statements that reflect trading for money as the purpose. These keywords include Chinese phrases such as “ying” (“win”) and “zhuanqian” (“making money”). Any simulated account with a description that contains such keywords is classified as a trading-for-performance account. Using this method we classify 8,445 accounts

B. Watch list

As discussed in Section 3.1, some investors may use simulated accounts as “watch lists.” As discussed in Section 3.1, fewer than 17% of real account purchases are suspected of being driven by the watch-list motivation, so we view it as unlikely that the watch-list hypothesis explains the purpose for which Snowball investors use their simulated accounts. Nevertheless, in Appendix Section B, we conduct a robustness test in which we exclude investors who are suspected of using their simulated accounts as watch lists. We define such investors as those who ever purchase, in our sample period, any stock for a real account when the stock was held in a simulated account at the time of the purchase. Such a definition is perhaps too broad, as it may exclude many false positives. We use this criterion to ensure that the remaining sample is free of the “watch-list” issue. Overall, the evidence is inconsistent with the “watch-list” hypothesis. The “watch-list” hypothesis also predicts that adding a simulated account should facilitate searching and stock selection and hence strengthen biases such as lottery preference and extrapolation. In Section 5.4, we show that the results are exactly the opposite to this prediction.

C. Time-series design

For the main tests, we restrict our sample to periods during which an investor holds both real and simulated accounts. Choosing the same period enables us to control for time-varying effects. The existence of real accounts may, however, indicate that investors are using their simulated accounts for purposes other than performance maximization. In Internet Appendix Section C, we report a time-series test comparing the two types of accounts across different periods. Specifically, we compare a real account with simulated trading before an investor opens a real account. Please note that this remains a within-subject test. Again, we find very similar results.

D. By the ratio of overlapped holdings

In Section 3.3, we observe that investors tend to hold the same stocks in both types of accounts, but the ratio of overlapped holdings varies significantly across investors. In this subsection,

as trading-for-performance accounts. The results based on this subsample are similar (Appendix Section A). The number of accounts classified as trading-for-performance accounts is 33% higher than that based on the manual method, a finding that is consistent with the conservatism we adopt in the manual classification process.

we report results by the ratio of the number of overlapped holdings to the total number of real account holdings.²⁷ Specifically, we split the sample into three subsamples. Investors with no overlapped holdings are classified as a separate group. The remaining investors are split into two equal-sized subsamples. The results presented in Internet Appendix Section D demonstrate that our results are robust across the overlapped-holdings subsamples. High values of the overlapping-holdings ratio may indicate that a similar thought process is associated with both types of accounts. The fact that the results are robust even when the ratio is very high indicates that the real–simulated differences likely reflect differences in stakes instead of other differences.

E. Between-subject design

Our within-subject design can identify the causal effects of stakes and carry strong statistical power. Experimental results may, however, be sensitive to whether a design is within-subject or between-subject (Charness, Gneezy, and Kuhn, 2012). For example, consider ambiguity aversion. In Fox and Tversky (1995), one set of gambles is associated with clear odds of winning while the other set of gambles is vague in this respect. The within-subject analysis (when individuals are given both clear and vague gambles) indicates ambiguity aversion, while the between-subject analysis does not. To assess whether our results are sensitive to the choice between a within- and a between-subject design, in Internet Appendix Section E, we calculate the biases for investors who hold only simulated accounts and for those who hold only real accounts. The results are similar to the baseline results.

F. Peer effect

Our data are drawn from an investor social network and our investors’ trades can be observed by their peers. Such observability may affect investor behaviors (Heimer, 2016). In Internet Appendix Section F, we examine whether such observability has any impact on stakes effects. Specifically, we replicate our baseline results by restricting our sample to investors who lack followers. Presumably, these investors are those least affected by the observability of trades. The results are similar to the baseline results.

²⁷ The results based on the ratio of the number of overlapped holdings to the total number of simulated account holdings are very similar.

G. Other robustness tests

First, in Internet Appendix Section G, we present robustness tests in which we use alternative measures of the behavioral biases and find qualitatively similar results. Second, in Internet Appendix Section H, we report the results for the disposition effect using a pooled regression approach. The individual-level disposition effect estimates may be noisy. The pooled regression results confirm that the disposition effect is stronger in real accounts. Third, as discussed in Section 3.3, among the 4,413 investors, 3,332 hold multiple simulated accounts. In Internet Appendix Section I, we replicate the results reported in Table 6 by restricting the sample to investors who hold only one simulated account and find similar patterns. Fourth, as discussed in Section 3.2, Snowball limits the total number of trades that are available on the website. We then consider the 19% of the sample whose trades are truncated on the Snowball website. We replicate Table 6 by restricting the sample to investors whose trades are not truncated or were executed in the final year of our sample, when such data truncation is least likely. The results, which are reported in Internet Appendix Section J, are similar to the main results.

Overall, the results reported in Table 6 and Sections A–J of the Internet Appendix indicate clearly that biases are stronger and performance is worse in real accounts than in simulated accounts. Such findings are inconsistent with the argument that higher stakes will weaken biases. In fact, they point in the opposite direction: biases are stronger when the stakes are higher.

5.2 Behavioral consistency between real and simulated accounts

The findings reported in Table 6 have strong implications for the experimental method as a means of understanding human behaviors. Consistent with many skeptical criticisms, our findings show that biases documented in low-stakes lab settings may differ from those that emerge in high-stakes real-world settings. Instead of observing that high stakes eliminate biases, however, as many skeptical scholars have argued, we observe that the low-stakes lab method may underestimate the biases harbored by real-world humans, at least in the case of stock markets. Can we learn anything, then, from lab methodology about real-world human behaviors?

For Table 7, we examine whether and how biases are correlated between high-stakes real accounts and low-stakes simulated accounts. Although the lab method may underestimate or

overestimate bias, it may still contain information about heterogeneity across investors. For example, if investor A holds a stronger lottery preference than investor B, she/he may exhibit a stronger lottery preference in both her/his real and simulated accounts than investor B, leading to positive correlations across investors between the real-account lottery preference and the simulated-account lottery preference.

This is indeed what we document. In the columns labeled “Correlations” and “p-values” we report the correlation coefficients of the biases between real accounts and simulated accounts as well as their p-values. It is evident that, between the same investors’ real and simulated accounts, the biases are strongly positively correlated. The correlation coefficients are all highly statistically significant, ranging from 0.132 for the disposition effect to 0.444 for local bias.

We then consider another method for testing the correlations. We first sort all investors into five quintiles based on the value of a given bias in their real accounts. For each quintile, we then calculate the average value of the bias for both their real accounts and their simulated accounts. We also report the means of and t-statistics for the differences between biases in the highest quintile and those in the lowest quintile. For all seven measures, we see that sorting by the value of a bias in real accounts leads to a sizable difference in the value of the bias in simulated accounts. These differences are wider than the interquartile ranges of these biases (see Table 4), except with respect to underdiversification and performance, for which the difference/interquartile range ratio is above 60%.

The results reported in Table 7 indicate strong behavioral consistency. These results also suggest that, even when there are low stakes in simulated accounts, investors are not trading randomly and lab experiments can still help researchers study behaviors.

5.3 Investor heterogeneity in stakes effects

Do some investors exhibit stronger stakes effects than others? We examine this possibility and report the results in Table 8. For each bias, we calculate the individual-level differences between the real and simulated accounts. In Table 8, we report the correlation coefficients between these differences across investors. We expect the correlations to be positive if the stakes effects have an investor-level component. In other words, if higher stakes increase a bias of a given investor to a

significant (small) extent, it may also increase other biases of this investor to a significant (small) extent.

The results reported in Table 8 show that, across all the bias measures, most of the correlation coefficients are positive and all of them are negatively correlated with performance. As is the case with other tables, here local bias is an exception: the real–simulated difference in local bias is not strongly correlated with the real–simulated differences in other biases or performance. The real–simulated difference in underdiversification is also negatively correlated with the real–simulated differences in lottery preferences and extrapolation, although this result is not statistically significant. Overall, the evidence suggests that there is an individual component of the stakes effects.

In Table 9, we report correlations between investor characteristics and their stakes effects. For each bias, the stakes effects are measured as real–simulated differences. We consider four investor characteristics: portfolio size (calculated based on the max-min rule method discussed at the end of Section 3), the number of followers on Snowball, the number of posts on Snowball, and GDP per capita in the investor’s city of residence. The results show that the stakes effects are weaker for investors who hold larger portfolios and investors based in relatively developed areas, while they are stronger for investors who are more active on social media as measured by the number of posts. The number of followers does not consistently correlate with the effects of the stakes.

The findings regarding the cross-investor heterogeneities in stakes effects shed valuable light on the mechanisms underlying those effects. Given the data, we have limited demographic information to analyze and await future studies to examine this question more thoroughly.

5.4 Stakes, attention, and adding a second type of account

An often implicit assumption in discussions of the stakes–behaviors relationship is that raising the stakes will lead to greater attention and greater effort. Recall our findings that investor turnover is higher in real accounts than in simulated accounts. Trading volume is a widely used proxy for attention (Barber and Odean, 2008). To the extent that trading volume is a valid proxy for investor attention, the results pertaining to turnover are consistent with the idea that higher stakes are associated with greater attention. In this section, we take advantage of the fact that the real and simulated accounts are not opened at the same time and present analyses to explore how adding a

second type of account affects investors' trading behavior in an existing account. Adding a second type of account may reduce investor attention to the existing account. If the stakes effects are driven partially by attention, we expect to find that adding a second type of account will reduce biases in the existing account if attention strengthens biases.

Specifically, we compare biases between the overlapping periods and the non-overlapping periods, i.e., periods when an investor has not opened a second type of account. Such an estimate remains a within-subject estimate but it is calculated across periods. Most (3,807 of 4,413) of the investors opened their simulated accounts first, with only 606 having opened their real accounts first. For the latter, most opened their real accounts not long before they opened their simulated accounts.

We exclude investors whose non-overlapping periods are shorter than 90 days. As is the case with our main results, such a requirement ensures that we can obtain meaningful estimates of the bias/performance measures. Differing from the main test in which we require a minimum 365-day window, however, here we use a 90-day window. Whether we require a 180-day or a 365-day window, the results derived by comparing the simulated accounts between non-overlapping and overlapping periods are similar. If we require a 180-day window, however, the sample for the real-account comparison will be roughly halved and become too small for a meaningful statistical comparison. With the 90-day requirement, we obtain 2,538 investors for the simulated-account comparison and 203 investors for the real-account comparison.

We report the results in Table 10. The results reported in Panel A enable us to compare the results for simulated accounts between non-overlapping and overlapping periods. The results reported in Panel B enable us to compare the results for real accounts. Maintaining an account of the other type is associated with significantly lower turnover in both simulated and real accounts, consistent with our conjecture that adding a second account distracts an investor's attention from existing accounts. Consistent with our hypothesis that increased attention strengthens biases, we find that maintaining an account of the other type is associated with weaker biases in both simulated accounts and real accounts. For simulated accounts, only the differences for those pertaining to local bias lack statistical significance. For real accounts, the differences are statistically significant only for lottery preference, underdiversification, and turnover, perhaps reflecting the smaller sample. The existence

of an account of the other type is associated with inferior portfolio performance for both simulated accounts and real accounts.

These results also shed doubt on the above-discussed mechanism whereby investors use simulated accounts as “watch lists.” If simulated accounts are used as “watch lists,” we expect to find that holding a simulated account strengthens the selection of lottery stocks and stocks that have earned high past returns, leading to stronger lottery preferences and extrapolation. The results reported in Panel B contradict these predictions.²⁸

5.5 Stakes, cognitive dissonance, and self-image

We conduct additional analyses of cognitive dissonance and self-image by taking advantage of the feature in virtue of which Snowball is a social platform that allows investors to write posts. Investors may find that admitting a mistake in a real account is much more embarrassing and wounding to one’s self-image than admitting a mistake in a simulated account. If cognitive dissonance and self-image affect investors, we expect them to post more often for winners than for losers. If investors care about real accounts more than they care about simulated accounts, we expect to find that the winner–loser difference in postings is greater for real accounts than for simulated accounts.

In Table 11, we report results pertaining to the probability that an investor writes a post within the next three days about a stock he/she has just traded.²⁹ We report the results for buying and selling separately. With real accounts the probabilities of posting are 3.74% and 2.41% for buying and selling, respectively. With simulated accounts the probabilities are 1.14% and 1.08%, respectively. The real–simulated differences are highly statistically significant.

In the next three rows of Table 11, we report the probability of posting conditional on selling a winner, selling a loser, and winner–loser differences. Investors are significantly more likely to

²⁸ These results also shed light on the co-existence effect discussed in Section 5.1.2. In our main tests, we restrict our sample to periods during which an investor holds both real and simulated accounts. This design provides within-subject and within-the-same-period estimates of stakes effects. Our within-subject estimates avoid the impact of investor heterogeneities. Choosing the same period enables us to control for time-varying effects. The co-existence of both types of accounts may also, however, affect investor behaviors. For example, evidence shows that ambiguity aversion exists mainly when a more ambiguous choice and a less ambiguous choice co-exist (Fox and Tversky, 1995). Recall that our results are robust to a between-subject design (see Internet Appendix Section E). The results presented in this section also suggest that the co-existence of both types of accounts does not drive the results.

²⁹ The results are similar if we vary the window for measuring posting activities to one day or seven days. Appendix Section K presents these robustness results.

broadcast their investment successes than their failures. The winner–loser differences are 1.33% and 0.53% for real and simulated accounts, respectively. The positive winner–loser differences are consistent with the presence of self-enhancing transmission bias (Hirshleifer, 2020; Han, Hirshleifer, and Walden, 2022; Sui and Wang, 2025). Interestingly, the difference in winner–loser postings is greater in the real accounts than in the simulated accounts, suggesting the presence of a stronger self-enhancing transmission bias when trading for real money. Overall, these results confirm the prediction that stakes strengthen the effects of cognitive dissonance and self-image.

6. Results based on two brokerage datasets

In this section, we present analyses using two brokerage account datasets. One motivation for this analysis is to provide external validity. There are concerns regarding whether our results can be generalized to other settings to mitigate the concern that accounts carrying no monetary stakes may constitute a special case. Conceptually, the stakes effect does not have to be monotonic. The marginal effect of stakes on performance may be negative when stakes are zero or very low if raising the stakes reduces decision noise and leads investors with incorrect beliefs or nontraditional preferences to make decisions that conform more closely to how they think and what they prefer. When stakes are sufficiently high, the marginal effect may become positive if additional effort shifts investors from the use of heuristics to a more deliberative approach to processing. Although such a non-monotonicity possibility does not invalidate our identification’s internal validity, it raises questions about the study’s external validity, i.e., whether we can generalize the results to other settings. The analysis in this section addresses this concern.

The second motivation of the brokerage account analysis is to examine another bias—the reverse disposition effects of mutual funds. Chang, Solomon, and Westerfield (2016) document that when investors trade actively managed mutual funds, they exhibit a reverse disposition effect instead of the disposition effect and explain the findings based on cognitive dissonance. The Snowball data include very few actively managed funds.³⁰ The brokerage dataset allows us to examine the reverse disposition effect.

³⁰ Snowball covers only exchange-traded securities. The funds it covers include exchange-traded funds (ETFs), listed open-end funds (LOFs), and closed-end funds, most of which are passive index funds. As argued by Chang, Solomon,

We collect one dataset from a large US discount brokerage company that has been widely used in retail investor studies (Barber and Odean, 2000). The raw data include trading activity for roughly 78,000 households between January 1991 and November 1996. We collect a similar dataset from a large Chinese brokerage company for a sample period that runs from 2000 through 2009 (Frydman and Wang, 2020; An et al., 2024). For the reverse disposition effect analysis, we can use only the US dataset because the Chinese brokerage dataset does not include non-stock trades.

6.1 The disposition effect in stocks

In the setting, we measure stakes by a position’s portfolio weight and compare the disposition effect between positions. If stakes strengthen investor biases, we expect to find a stronger disposition effect on stocks allocated higher portfolio weights. The unit of observation is an investor-stock-day triple. Please see An et al. (2024) for details regarding portfolio construction. As our empirical model, we use Equation (2),

$$Sell_{i,j,t} = \beta_1 Gain_{i,j,t-1} + \beta_2 Weight_{i,j,t-1} + \beta_3 Gain_{i,j,t-1} * Weight_{i,j,t-1} + controls + \varepsilon_{i,j,t}, \quad (2)$$

where $Sell_{i,j,t}$ and $Gain_{i,j,t-1}$ are defined in the same way as in Section 4.1.1. $Weight_{i,j,t-1}$ is position j ’s value as a percentage of total portfolio value at the end of day $t-1$. The value of position j at the end of day $t-1$, if measured using the price at $t-1$, will be mechanically correlated with the $Gain$ dummy. To avoid such confoundedness, we measure the value of a position at the end of day $t-1$ as the product of the number of shares at $t-1$ and the *purchase* price instead of the price at $t-1$.³¹ We exclude investor-days with one only stock in a portfolio.

We report the results in Table 12. For both US data (Panel A) and Chinese data (Panel B), we estimate Equation (2) in five distinct subsamples based on the number of holdings. We conduct the analysis by reference to numbers of holdings because the number of holdings is mechanically

and Westerfield (2016), an index fund’s fund manager may not be a credible target to blame for the fund’s poor performance. Of our 4,413 investors, only 13 have traded actively managed funds (defined as general equity funds, balanced funds, or flexible allocation funds) in both real and simulated accounts.

³¹ Calculating *PortWeight* using the Snowball data is difficult. As discussed in Section 3.2, we have data only on the fraction of trades at trade time, not at the end of every day. Inferring the portfolio weights at the end of each day requires information on stock price changes since the last trade as well as cash inflows and outflows. Unfortunately, data on cash outflows and inflows are not available. Nevertheless, in Appendix Section L, we measure *PortWeight* with the portfolio weight at the time a position was established. In the Snowball data, the coefficients on $Gain * PortWeight$ are positive and statistically significant in most cases, which is consistent with the results based on the two brokerage company datasets.

correlated with *PortWeight*. For example, the average *PortWeight* is 0.5 (0.25) when the number of holdings is 2 (4).³² In all regressions we include investor fixed effects, stock fixed effects, and date fixed effects. In all subsamples, the coefficients on *Gain*PortWeight* are positive and highly statistically significant. The lowest *t*-statistic is above 5. The economic effect is also sizable. Based on the estimates reported in column (5), when *PortWeight* increases from 25% to 75%, the disposition effect increases from 0.23 to 0.48 in the US data and from 3.61 to 5.81 in the Chinese data. These changes represent a 107% increase and a 61% increase, respectively.

6.2 The reverse disposition effect in equity mutual funds

If stakes strengthen investor biases, we expect to find a stronger reverse disposition effect on equity mutual funds with greater portfolio weights. As in the disposition effect analysis, here we use Equation (2) as our empirical model. As with stocks, an investor's purchase price for a fund is calculated as the volume-weighted average price using the trades file. Daily price information for mutual funds is not available during the sample period. Hence, following Chang, Solomon, and Westerfield (2016), we use the most recent trade price from the trades file and define the variable *Gain* as equal to one if this most recent price is greater than an investor purchase price and zero otherwise. The results reported in Table 13 show that the coefficients on *Gain*PortWeight* are negative in all subsamples and are statistically significant in four cases. Based on the estimates reported in column (5), when *PortWeight* increases from 25% to 75%, the reverse disposition effect changes from -0.13 to -0.26, a sizable 100% change.

The results are robust if we use Chang, Solomon, and Westerfield's (2016) specifications. Chang, Solomon, and Westerfield (2016) conduct the analysis only with investor-months when an investor sold some position in their overall portfolios. The results, reported in Internet Appendix Section M, are robust to this alternative specification.

³² Chang, Solomon, and Westerfield (2016) find that the disposition effect in stocks and the reverse disposition effect in funds are both greater in magnitude for investors who hold fewer assets. This finding is consistent with our findings, as holding fewer assets is positively associated with a higher average value of *PortWeight*. We conduct our analysis based on the number of assets, showing that the portfolio weight effect is robust even after controlling for the effect of the number of assets.

6.3 Discussion

Overall, these results show that investors exhibit a stronger disposition effect with regard to stocks that are allocated higher portfolio weights and they exhibit a stronger reverse disposition effect with regard to funds that are allocated higher portfolio weights. Both sets of results are consistent with our results based on Snowball data, indicating that higher stakes are associated with stronger biases. The results based on the two brokerage datasets provide external validity, showing that the higher-stakes-stronger-biases results are not restricted to data from the online investor platform or high-monetary and no-monetary comparisons. The findings pertaining to the reverse disposition effect also provide additional evidence supporting the stakes effect and the cognitive dissonance mechanism.

7. Conclusion

Our study's headline finding is that investors exhibit stronger biases and worse performance in higher-stakes real accounts than in lower-stakes simulated accounts. The results hold for both a within-subject design and a between-subject design. These results support the notion that stakes affect behaviors. In contrast to the conventional wisdom that higher stakes should weaken biases, our evidence suggests that higher stakes can strengthen biases. We also find that investors exhibit strong biases and their simulated accounts perform poorly even though these accounts do not involve real money. Across investors, biases are strongly correlated between the two types of accounts, suggesting behavioral consistency. These findings suggest that low-stakes lab methods, although imperfect, can still be informative about real-world human behaviors.

Our results indicate that attention should be considered when examining stakes effects. Evidence from investor trading shows that investors pay greater attention to their real accounts than their simulated accounts. Opening a real (simulated) account predicts the weakening of biases in an existing simulated (real) account, perhaps because attention is partially reallocated from the existing account to the new account.

The expectation that higher stakes will affect behaviors rests on two assumptions. First, raising the stakes leads to greater attention and effort. Second, such an increase in attention and effort can

affect behaviors. Overall, our findings imply that, in the stock market, the marginal effect of attention is negative. Perhaps investors carry incorrect mental models and greater attention impairs decision-making (Dawes, 1979; Dawes, Faust, and Meehl, 1989). It is also possible that investors have non-traditional preferences and that higher stakes lead to behaviors that conform more closely to investors' preferences but deviate from traditional benchmarks. Higher stakes can also strengthen the elements of non-traditional preferences, such as prospect theory and cognitive dissonance. We report two findings that are consistent with cognitive dissonance and self-image. First, we find a stronger reverse disposition effect on equity mutual funds that are allocated higher portfolio weights. Second, investors are more likely to broadcast their selling of winners than their selling of losers. Such a winner–loser difference is more pronounced with real accounts than with simulated accounts.

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Figure 1. Graphic views of differences between real and simulated accounts

This figure presents the estimated kernel densities for behavioral biases in real and simulated accounts. The red lines represent real accounts and the blue lines represent simulated accounts. Each plot uses a Gaussian kernel. The x-axis for turnover is the log scale.

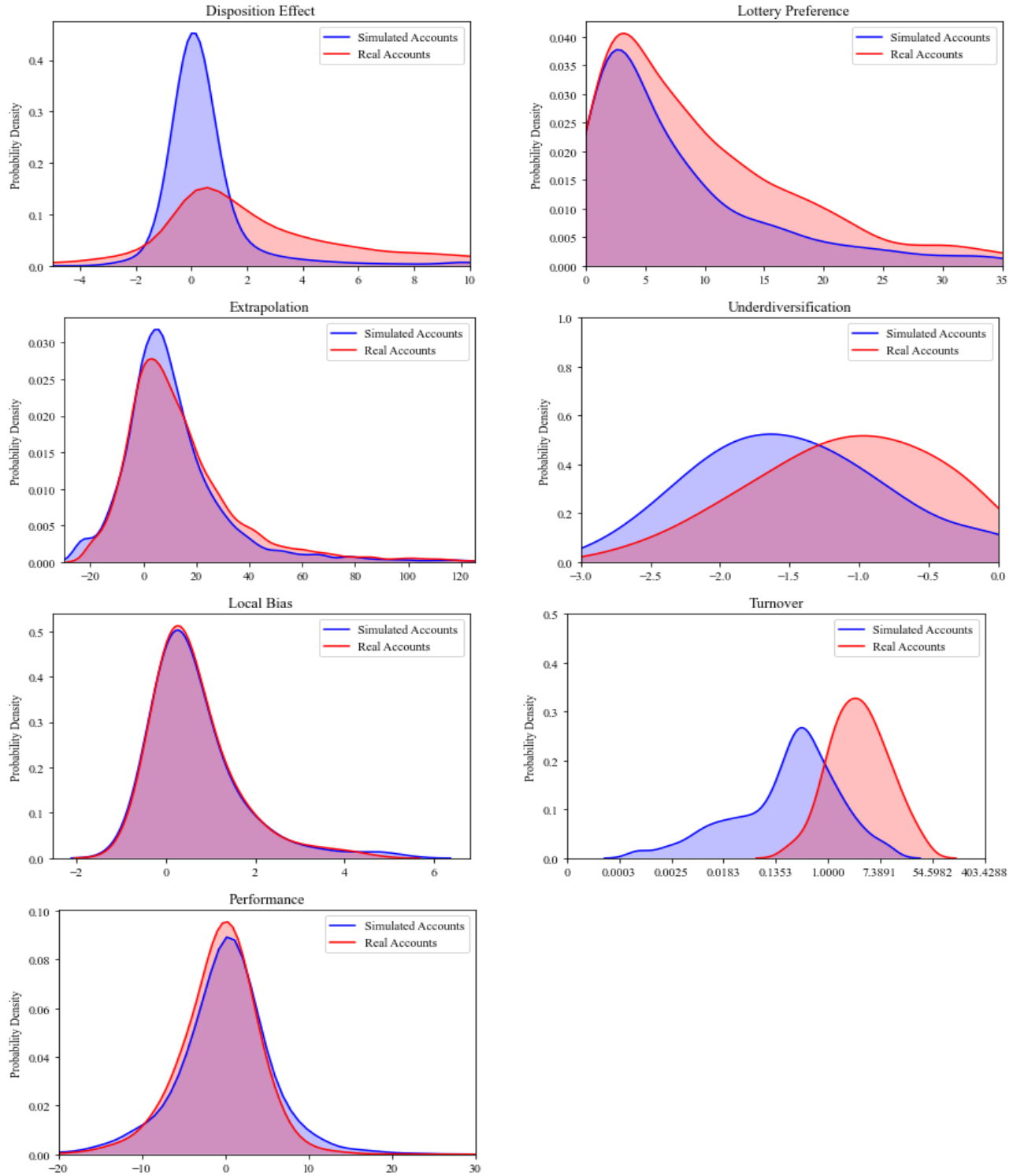


Table 1. Literature on the effects of stakes

This table summarizes studies on the effects of stakes. We classify these studies into five groups (and eight subgroups) and list the relevant tasks and representative studies.

Stakes effects	Tasks and representative studies
Higher stakes improve performance.	A. Memory or recall tasks and clerical tasks (e.g., coding words or building things) (Camerer and Hogarth, 1999); the Cognitive Reflection Test (Enke et al., 2021); academic performance (Levitt et al., 2016)
Higher stakes do not have significant effects.	B. Tasks involving high cognitive effort costs, such as tasks testing base rate neglect, anchoring, and failure of contingent thinking (Enke et al., 2021)
Higher stakes can hurt performance.	C1. Tasks involving creativity or tasks in which simple intuition or habit provides optimal answers and thinking harder makes things worse (Arkes, Dawes, and Christensen, 1986; Ashton, 1990; Hogarth et al., 1991; Ariely et al., 2009). C2. Tasks in which subjects form incorrect mental models, such as clinical judgments (Dawes, Faust, and Meehl, 1989), predictions of faculty ratings of graduate students (Dawes, 1971), product valuations plus learning from new information and the formation of stock market expectations (Hartzmark, Hirshman, and Imas, 2021)
Higher stakes reduce the weight of nonpecuniary factors in decision-making.	D1. The ultimatum game (Slonim and Roth, 1998; Cameron, 1999; Andersen et al., 2011); coordination games (Parravano and Poulsen, 2015); a three-person centipede game (Rapoport et al., 2003); gaming against managers (Cooper et al., 1999). D2. Extrinsic monetary incentives crowd out intrinsic incentives such as volunteering and other pro-social behaviors (Gneezy and Rustichini, 2000; Gneezy et al., 2011; Bowles and Polania-Reyes, 2012).
Higher stakes change risk preferences.	E1. Choice of risky gambles (Kachelmeier and Shehata, 1992; Tversky and Kahneman, 1992; Holt and Laury, 2002, 2005; Hershey and Schoemaker, 1980; Scholten and Read, 2014; Fehr-Duda et al., 2010; Khaw, Li, and Woodford, 2022) E2. Choice of risky gambles involving features of affect or emotion (Rottenstreich and Hsee, 2001; Hsee and Rottenstreich, 2004; Faro and Rottenstreich, 2006; Andersson et al., 2016)

Table 2. Stakes and each bias

This table summarizes prominent interpretations in the literature of the studied biases and the roles that stakes may play. See Section 2 for a fuller discussion.

	Prominent Interpretation	The Role of Stakes
Disposition effect	An incorrect belief in the mean reversion of prices (Odean, 1998) The diminishing sensitivity of prospect theory (Shefrin and Statman, 1985; Barberis and Xiong, 2009) Cognitive dissonance (Chang, Solomon, and Westerfield, 2016) Realization utility (Barberis and Xiong, 2012; Ingersoll and Jin, 2013)	Stakes lead to more careful application of an incorrect belief. Stakes lead to a more pronounced value function. Stakes make cognitive dissonance more painful. Stakes make gain (loss) realization more joyful (painful).
Lottery preference	Probability weighting function of prospect theory (Barberis and Huang, 2008) Salience theory (Bordalo, Gennaioli, and Shleifer, 2012)	Stakes lead to a more pronounced probability weighting function. Stakes increase the salience of a jackpot.
Extrapolation	Representativeness heuristics (Kahneman and Tversky, 1972; Barberis, Shleifer, and Vishny, 1998) Belief in the “law of small numbers” (Tversky and Kahneman, 1971)	Stakes lead to more careful application of an incorrect belief. Stakes lead to more careful application of an incorrect belief.
Underdiversification	Ambiguity aversion Mere exposure effect (Zajonc, 1968; Bornstein, 1989) Probability weighting (Barberis and Huang, 2008)	Stakes increase the disutility of unknown risks. Stakes may not affect the mere exposure effect. Stakes lead to more pronounced probability weighting and investor sacrifice diversification for skewness exposure.
Local Bias	Ambiguity aversion Mere exposure effect	Stakes increase the disutility of unknown risks. Stakes may not affect the mere exposure effect.
Turnover	Overconfidence (Odean, 1998; Barber and Odean, 2000; Scheinkman and Xiong, 2003) Sensation-seeking (Grinblatt and Keloharju, 2009)	Stakes lead to more careful application of an incorrect belief. Stakes make decisions more sensational.

Table 3. Summary Statistics

In this table, we report summary statistics for the sample investors. These investors include all those whose real accounts and simulated accounts overlap for more than a year. We report summary statistics for these investors for the overlapping period. *Active period* is the length of an active period in days. *Number of trades* is the total number of trades in the overlapping period. Overlapped holdings are the number of stock-days on which an investor holds a stock in both types of accounts on the same day. Real holdings (simulated holdings) are the number of stock-days in real (simulated) accounts. We report the total numbers (N), means, standard deviations (STD), first quartiles (Q1), medians, and third quartiles (Q3).

	N	Mean	STD	Q1	Median	Q3
Active period	4,413	667.867	197.555	491.000	642.000	833.000
Number of trades in real accounts	4,413	99.661	62.729	43.000	91.000	160.000
Number of trades in simulated accounts	4,413	39.772	158.106	6.429	14.000	33.667
Overlapped holdings/Real holdings	4,413	0.356	0.315	0.049	0.298	0.602
Overlapped holding/Simulated holdings	4,413	0.149	0.196	0.011	0.072	0.208

Table 4. Summary statistics for biases

In this table, we report the mean, standard deviation (STD), Q1 (the first quartile), median, and Q3 (the third quartile) for each type of bias. We also report t -statistics for and Wilcoxon p -values of the investor-level bias measures. See Section 4.1 for the bias definitions. All biases are calculated at the individual investor level and separately for real accounts and simulated accounts. We winsorize each measure separately for the real and simulated accounts at the 1% and 99% levels.

Panel A. Real accounts

	N	Mean	STD	Q1	Median	Q3	t	Wilcoxon p
Disposition effect	4,371	4.163	7.532	0.160	1.868	5.908	(36.54)	0.000
Lottery preference	4,369	6.491	9.130	0.000	2.495	9.708	(46.99)	0.000
Extrapolation	4,373	15.599	26.191	0.121	9.224	22.374	(39.39)	0.000
Underdiversification	4,394	-1.086	0.657	-1.536	-1.041	-0.580	(-109.61)	0.000
Local Bias	1,739	0.630	0.880	0.055	0.366	0.907	(29.84)	0.000
Turnover	4,386	5.960	7.382	1.501	3.180	7.166	(53.47)	0.000
Performance	4,375	-0.656	4.685	-3.181	-0.285	2.174	(-9.27)	0.000

Panel B. Simulated accounts

	N	Mean	STD	Q1	Median	Q3	t	Wilcoxon p
Disposition effect	4,371	0.489	1.575	0.000	0.000	0.353	(20.54)	0.000
Lottery preference	4,369	4.264	8.792	0.000	0.068	4.441	(32.06)	0.000
Extrapolation	4,373	12.660	25.200	-0.199	7.403	18.028	(33.22)	0.000
Underdiversification	4,394	-1.473	0.688	-1.967	-1.539	-1.073	(-141.96)	0.000
Local Bias	1,739	0.645	0.970	0.048	0.354	0.914	(27.74)	0.000
Turnover	4,386	0.839	1.624	0.070	0.304	0.784	(34.22)	0.000
Performance	4,375	-0.148	5.921	-2.858	0.219	2.936	(-1.65)	0.020

Table 5. Correlation coefficients of biases

In this table, we report pairwise correlation coefficients between behavioral biases: Panel A lists the correlation coefficients of real accounts and Panel B lists the correlation coefficients of simulated accounts. See Section 4.1 for the bias definitions. Reported in parentheses are the p -values of the correlation coefficients.

Panel A. Real accounts

	Disposition effect	Lottery preference	Extrapolation	Underdiversification	Local bias	Turnover	Performance
Disposition effect	1						
Lottery preference	0.213 (0.000)	1					
Extrapolation	0.115 (0.000)	0.157 (0.000)	1				
Underdiversification	0.237 (0.000)	0.211 (0.000)	0.117 (0.000)	1			
Local bias	0.050 (0.039)	-0.068 (0.005)	-0.044 (0.067)	-0.007 (0.774)	1		
Turnover	0.306 (0.000)	0.234 (0.000)	0.269 (0.000)	0.368 (0.000)	-0.021 (0.390)	1	
Performance	-0.209 (0.000)	-0.260 (0.000)	-0.259 (0.000)	-0.180 (0.000)	-0.003 (0.915)	-0.229 (0.000)	1

Panel B. Simulated accounts

	Disposition effect	Lottery preference	Extrapolation	Underdiversification	Local bias	Turnover	Performance
Disposition effect	1						
Lottery preference	0.119 (0.000)	1					
Extrapolation	0.137 (0.000)	0.045 (0.003)	1				
Underdiversification	0.104 (0.000)	0.063 (0.000)	0.042 (0.006)	1			
Local bias	-0.028 (0.250)	0.036 (0.131)	-0.030 (0.209)	0.048 (0.045)	1		
Turnover	0.413 (0.000)	0.124 (0.000)	0.230 (0.000)	0.052 (0.001)	-0.027 (0.268)	1	
Performance	-0.093 (0.000)	-0.099 (0.000)	-0.129 (0.000)	-0.094 (0.000)	-0.041 (0.088)	-0.082 (0.000)	1

Table 6. Differences between the real and simulated accounts

In this table, we report differences between real and simulated accounts. See Section 4.1 for the bias definitions. For each type of bias we report the mean of the real accounts, the mean of the simulated accounts, and the mean of the differences between them. For the differences, we report t -statistics and Wilcoxon p -values. In column “N” we report the number of investors.

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	4,371	4.163	0.489	3.674	(32.43)	0.000
Lottery preference	4,369	6.491	4.264	2.228	(13.88)	0.000
Extrapolation	4,373	15.599	12.66	2.939	(6.49)	0.006
Underdiversification	4,394	-1.086	-1.473	0.386	(32.67)	0.000
Local Bias	1,739	0.630	0.645	-0.015	(-0.64)	0.810
Turnover	4,386	5.960	0.839	5.121	(47.44)	0.000
Performance	4,375	-0.656	-0.148	-0.509	(-4.95)	0.001

Table 7. Correlations between real and simulated accounts

In this table, we report results indicating how biases are correlated between real and simulated accounts. In the “Correlation” column, we report the correlation coefficients and their p -values between a type of bias in an investor’s real account and the same type of bias in her/his simulated account. In the next five columns, we report the mean of each bias measure by quintiles sorted by the value of the bias in an investor’s real account. In the High–Low column, we report the mean differences between the high and low groups as well as the t -statistics (in parentheses).

	Correlation		Low 1	2	3	4	High 5	High–Low	
	Coefficient	p						Mean	t -stat
Disposition effect	0.132	0.000							
Real accounts			-2.434	0.454	1.903	4.859	16.041	18.476	(62.00)
Simulated accounts			0.332	0.176	0.407	0.609	0.923	0.591	(6.62)
Lottery preference	0.300	0.000							
Real accounts			0.000	0.042	2.511	8.069	21.829	21.829	(75.26)
Simulated accounts			2.090	2.146	3.121	5.276	8.684	6.594	(14.03)
Extrapolation	0.320	0.000							
Real accounts			-7.638	1.964	9.338	19.306	55.014	62.652	(56.58)
Simulated accounts			3.478	7.507	11.035	16.170	25.110	21.631	(15.46)
Underdiversification	0.321	0.000							
Real accounts			-2.067	-1.432	-1.044	-0.671	-0.217	1.851	(168.35)
Simulated accounts			-1.797	-1.612	-1.421	-1.355	-1.177	0.620	(19.30)
Local bias	0.444	0.000							
Real accounts			-0.175	0.111	0.371	0.775	2.067	2.242	(48.03)
Simulated accounts			0.280	0.329	0.491	0.775	1.349	1.069	(13.21)
Turnover	0.251	0.000							
Real accounts			0.802	1.788	3.258	6.130	17.829	17.028	(57.03)
Simulated accounts			0.351	0.536	0.732	0.996	1.582	1.231	(13.89)
Performance	0.195	0.000							
Real accounts			-7.407	-2.524	-0.322	1.669	5.301	12.708	(87.78)
Simulated accounts			-2.010	-0.679	-0.268	0.727	1.491	3.501	(11.59)

Table 8. The effects of stakes across investors

For this table we first calculate the difference between real and simulated accounts for each bias. In this table, we report the correlation coefficients between these differences across investors. The *p*-values of the correlation coefficients are reported in parentheses.

	Disposition effect	Lottery preference	Extrapolation	Underdiversification	Local bias	Turnover	Performance
Disposition effect	1						
Lottery preference	0.126 (0.000)	1					
Extrapolation	0.131 (0.000)	0.089 (0.000)	1				
Underdiversification	0.064 (0.000)	-0.002 (0.897)	-0.016 (0.305)	1			
Local bias	0.013 (0.597)	0.016 (0.505)	-0.017 (0.471)	0.030 (0.218)	1		
Turnover	0.296 (0.000)	0.096 (0.000)	0.187 (0.000)	0.125 (0.000)	0.008 (0.728)	1	
Performance	-0.098 (0.000)	-0.077 (0.000)	-0.159 (0.000)	-0.034 (0.025)	-0.022 (0.367)	-0.086 (0.000)	1

Table 9. Investor characteristics and stakes effects

For this table we first calculate the difference between real and simulated accounts for each bias. In this table, we report the correlation coefficients between these differences and investor characteristics. GDP per capita is measured at the city level. We report p -values of the correlation coefficients in parentheses.

	Log(Portfolio size)	Log(1 + Number of followers)	Log(1 + Number of posts)	Log(GDP per capita)
Disposition effect	-0.071 (0.000)	-0.032 (0.145)	0.068 (0.000)	-0.025 (0.084)
Lottery preference	-0.028 (0.029)	-0.013 (0.411)	0.007 (0.169)	-0.077 (0.004)
Extrapolation	-0.007 (0.410)	0.044 (0.086)	0.009 (0.996)	-0.004 (0.353)
Underdiversification	-0.124 (0.000)	0.007 (0.802)	0.042 (0.022)	-0.041 (0.058)
Local Bias	0.017 (0.791)	-0.051 (0.288)	-0.040 (0.104)	-0.040 (0.180)
Turnover	-0.084 (0.000)	-0.028 (0.205)	0.106 (0.000)	-0.055 (0.017)
Performance	0.038 (0.009)	0.000 (0.349)	-0.031 (0.228)	0.019 (0.687)

Table 10. Adding a second type of account

In this table, we report the effects of adding a second type of account on investor biases in existing accounts. For Panel A, we analyze biases in simulated accounts: before adding real accounts and after adding real accounts. For Panel B, we analyze biases in real accounts: before adding simulated accounts and after adding simulated accounts. We require the before periods to last at least 90 days.

Panel A. Simulated accounts

	N	Non-Overlapping: Before adding real accounts	Overlapping: After adding real accounts	Difference		
		Mean	Mean	Mean	<i>t</i> -stat	Wilcoxon <i>p</i>
Disposition effect	2,491	0.573	0.446	0.127	(2.33)	0.002
Lottery preference	2,422	4.466	3.853	0.613	(2.80)	0.302
Extrapolation	2,429	16.155	10.348	5.806	(8.69)	0.000
Underdiversification	2,505	-1.411	-1.495	0.084	(9.11)	0.000
Local Bias	1,068	0.617	0.625	-0.008	(-0.26)	0.168
Turnover	2,443	2.326	0.846	1.481	(17.18)	0.000
Performance	2,435	0.369	0.004	0.365	(2.05)	0.209

Panel B. Real accounts

	N	Non-Overlapping: Before adding simulated accounts	Overlapping: After adding simulated accounts	Difference		
		Mean	Mean	Mean	<i>t</i> -stat	Wilcoxon <i>p</i>
Disposition effect	205	2.650	2.294	0.356	(0.47)	0.141
Lottery preference	206	7.384	5.157	2.228	(2.15)	0.187
Extrapolation	206	15.473	13.851	1.622	(0.80)	0.444
Underdiversification	208	-0.886	-1.188	0.302	(7.63)	0.000
Local Bias	58	0.746	0.699	0.047	(0.36)	0.694
Turnover	208	7.640	3.599	4.041	(6.25)	0.000
Performance	206	0.732	-0.602	1.334	(2.31)	0.043

Table 11. Postings

In this table, we report results indicating the likelihood (in %) that an investor writes posts about his/her transaction on Snowball in the three days following a transaction. Each row corresponds to a distinct transaction type: buying, selling, selling winners, and selling losers. In the last row, we also report the difference between selling winners and selling losers. The analyses for real and simulated accounts are conducted separately. For each type of transaction we report the mean value of real accounts, the mean value of simulated accounts, and the mean value of the differences between them. For the differences, we report *t*-statistics and Wilcoxon *p*-values. In column “N,” we report the number of investors.

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	<i>t</i> -stat	Wilcoxon <i>p</i>
Buying	4,409	3.741	1.140	2.607	(21.76)	0.000
Selling	4,400	2.413	1.084	1.577	(10.64)	0.000
Selling winners	4,337	3.093	1.271	2.203	(11.32)	0.000
Selling losers	4,306	1.763	0.724	1.303	(8.61)	0.000
Selling winners – Selling losers	4,243	1.332	0.531	0.948	(4.96)	0.000

Table 12. The disposition effect in stocks based on two brokerage data sets

This table presents the pooled regression results of the disposition effect analysis. Specifically, we report the regression results obtained with the following model by the number of stocks in an investor's portfolio,

$$Sell_{i,j,t} = \beta_1 Gain_{i,j,t-1} + \beta_2 PortWeight_{i,j,t-1} + \beta_3 Gain_{i,j,t-1} * PortWeight_{i,j,t-1} + controls + \varepsilon_{i,j,t}$$

where observations occur at the investor (i), position (j), and date (t) levels. $Sell_{i,j,t}$ is a dummy variable that equals 1 if investor i sells position j (partially or fully) on day t . $Gain_{i,j,t-1}$ is a dummy variable that equals 1 if investor i experiences a gain on stock j at the end of day $t-1$. $PortWeight_{i,j,t-1}$ is the portfolio weight of position j in investor i 's portfolio on day $t-1$. We measure the value of a position using the purchase price of a stock instead of the current price. We add three control variables, following Ben-David and Hirshleifer (2012): the log purchase price, the square root of the number of days since purchase, and stock volatility calculated using daily returns during the 250 days preceding a purchase. For ease of presentation the coefficients are multiplied by 100. We cluster standard errors at the investor, stock, and day levels. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A. US

	(1) #Stocks=2	(2) #Stocks=3	(3) #Stocks=4	(4) #Stocks=5	(5) #Stocks>5
Gain	0.171*** (15.36)	0.164*** (16.09)	0.168*** (15.93)	0.169*** (14.32)	0.106*** (9.01)
PortWeight	0.030** (2.24)	0.036* (1.86)	0.083*** (3.20)	0.092*** (2.98)	-0.040 (-1.02)
Gain*PortWeight	0.184*** (8.59)	0.218*** (7.14)	0.200*** (5.53)	0.269*** (5.61)	0.493*** (8.40)
Log(Price)	0.027*** (2.74)	0.042*** (4.85)	0.036*** (3.75)	0.032*** (2.90)	0.035*** (3.81)
Sqrt(Time owned)	0.005*** (8.61)	0.002*** (4.26)	0.001 (1.60)	0.001 (1.49)	-0.002*** (-3.13)
Volatility	0.202 (0.55)	0.253 (0.71)	0.387 (0.97)	0.458 (1.04)	-0.170 (-0.64)
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Adj-R ²	0.027	0.027	0.027	0.026	0.012
Obs.	26,617,606	20,301,759	15,062,752	10,987,406	45,530,542

Panel B. China

	(1) #Stocks=2	(2) #Stocks=3	(3) #Stocks=4	(4) #Stocks=5	(5) #Stocks>5
Gain	3.225*** (51.99)	3.126*** (51.40)	3.070*** (51.75)	3.007*** (47.17)	2.514*** (38.58)
PortWeight	0.694*** (20.90)	1.102*** (23.01)	1.531*** (23.92)	2.018*** (22.42)	3.880*** (23.78)
Gain*PortWeight	0.562*** (7.42)	0.767*** (6.45)	1.061*** (6.30)	1.407*** (5.99)	4.395*** (12.30)
Log(Price)	0.790*** (17.56)	0.668*** (15.13)	0.587*** (13.37)	0.562*** (13.01)	0.559*** (12.88)
Sqrt(Time owned)	-0.169*** (-51.76)	-0.157*** (-48.90)	-0.153*** (-45.53)	-0.152*** (-42.25)	-0.157*** (-37.83)
Volatility	0.635** (2.45)	0.109 (0.50)	-0.093 (-0.37)	-0.121 (-0.44)	0.205 (0.86)
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Adj-R ²	0.078	0.073	0.071	0.070	0.053
Obs.	21,425,932	17,241,913	12,422,938	8,593,206	20,759,206

Table 13. The reverse disposition effect in funds based on the US brokerage dataset

This table presents the pooled regression results of the reverse disposition effect analysis. Specifically, we report the regression results obtained with the following model by the number of funds in an investor's portfolio,

$$Sell_{i,j,t} = \beta_1 Gain_{i,j,t-1} + \beta_2 PortWeight_{i,j,t-1} + \beta_3 Gain_{i,j,t-1} * PortWeight_{i,j,t-1} + controls + \varepsilon_{i,j,t}$$

where observations occur at the investor (i), position (j), and date (t) levels. $Sell_{i,j,t}$ is a dummy variable that equals 1 if investor i sells position j (partially or fully) on day t . $Gain_{i,j,t-1}$ is a dummy variable that equals 1 if investor i experiences a gain on fund j at the end of day $t-1$. $PortWeight_{i,j,t-1}$ is the portfolio weight of position j in investor i 's portfolio on day $t-1$. We measure the value of a position using the purchase price of a stock instead of the current price. We add two control variables, following Ben-David and Hirshleifer (2012): the log purchase price and the square root of the number of days since purchase. For ease of presentation the coefficients are multiplied by 100. We cluster standard errors at the investor, stock, and day levels. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1) #Funds=2	(2) #Funds=3	(3) #Funds=4	(4) #Funds=5	(5) #Funds>5
Gain	-0.014 (-1.33)	-0.056*** (-4.30)	-0.034** (-2.57)	-0.045*** (-2.71)	-0.062*** (-5.53)
PortWeight	0.160*** (4.68)	0.096* (1.82)	0.223*** (3.10)	0.437*** (4.03)	0.501*** (3.63)
Gain*PortWeight	-0.113*** (-3.18)	-0.053 (-1.02)	-0.185** (-2.54)	-0.211* (-1.93)	-0.270** (-1.97)
Log(Price)	0.246*** (6.58)	0.256*** (7.26)	0.235*** (5.31)	0.159*** (4.25)	0.078 (1.22)
Sqrt(Time Owned)	0.014*** (15.03)	0.013*** (13.98)	0.013*** (15.25)	0.012*** (14.12)	0.009*** (11.38)
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Adj-R ²	0.018	0.016	0.016	0.016	0.010
Obs.	7,766,602	6,368,504	4,936,451	3,621,131	10,809,225

Internet Appendix for
“Stakes and Investor Behaviors”

Pengfei Sui and Baolian Wang

Section A. Using simulated accounts that explicitly identify performance maximization as the purpose

In Table A1 we report results based on the subsample of investors who state explicitly that they seek to maximize simulated-account performance. For Panel A we identify these accounts manually. For Panel B we adopt keyword-based identification. If an account uses a performance-related keyword, we classify it as designed to maximize performance. The results are similar to our baseline results reported in Table 6.

Section B. Using simulated accounts as “watch lists”

As discussed in Section 3.1, some investors use simulated accounts as “watch lists” to keep track of stocks they are interested in buying but do not yet have strong enough conviction to buy. In this section, we first discuss the predictions associated with such a possibility and then present empirical tests.

Perhaps stocks with past returns or lottery features above certain thresholds are included in an investor’s real account, while stocks that fall just short are held in her simulated accounts. This would explain the real–simulated differences in extrapolation and lottery preferences. Investors may hold more stocks on a “watch list” than they will buy, which explains the real–simulated difference in underdiversification. Such an interpretation is, however, at odds with the results for local bias. If investors behave in a relatively explorative way when managing their simulated accounts, we predict that local bias will be lower in their simulated accounts than in their real accounts.

For Table A2 we conduct a robustness test in which we exclude investors who are suspected of using their simulated accounts as watch lists. We define such investors as those who ever purchase, in our sample period, any stock for a real account when the stock was held in a

simulated account at the time of the purchase. Such a definition is perhaps too broad, as it may include many false positives. We use this criterion to ensure that the remaining sample is free of the “watch-list” issue. The results reported in Table A2 lead to the same conclusion as those reported in Table 6. In fact, the differences between real and simulated accounts are greater in this subsample than those indicated in Table 6.

The “watch-list” hypothesis also predicts that adding a simulated account facilitates searching and stock selection and strengthens biases such as lottery preferences and extrapolation. In Section 5.4, we show that the results imply precisely the opposite conclusions.

Section C. Time-series designs

In Table A3, we report the results of a time-series test in which we compare the two types of accounts across periods. Specifically, we compare a real account with simulated trading before an investor opens a real account. Please note that this remains a within-subject test. Again, we find very similar results.

Section D. By the ratio of overlapped holdings

In Table A4, we report results obtained based on the ratio of the number of overlapped holdings to the total number of real account holdings. Specifically, we split the sample into three subsamples. Investors with no overlapped holdings are classified as a separate group. The remaining investors are split into two equal-sized subsamples. Our results are similar across overlapped holdings subsamples.

Section E. Between-subject designs

In Table A5, we examine stakes effects using a between-subject design. For real accounts we focus on users who do not hold any simulated accounts. For simulated accounts we use the period before an investor opens his/her real account. We require the account to be held for at least

one year. Another difference between this table and Table 6 is that, for Table 6, we conduct paired statistical tests for the differences while, for this table, the tests are not paired. The results are qualitatively similar to the results reported in Table 6.

Section F. Peer effect

Our investors' trades are observable by peers on Snowball. We examine whether such observability has any effect on the results. Specifically, we replicate our baseline results by restricting the analysis to investors who do not have any followers. Presumably, these investors are those least affected by the observability of trades. The results reported in Table A6 are similar to the baseline results.

Section G. Alternative measures

In this section, we introduce alternative measures of the biases and, in Table A7, we report the results of robustness tests based on these alternative measures. The Table A7 results are qualitatively similar to those reported in Table 6: relative to what occurs with simulated accounts, behavioral biases (except for local bias) are stronger in real accounts and performance is worse in real accounts.

A. The disposition effect

The first alternative measure of the disposition effect is that of Odean (1998). Specifically, we focus on investor-days when an investor sells at least one stock and has at least two stocks available for sale. We then calculate the propensity to sell a winning stock (PGR) and the propensity to sell a losing stock (PLR). The difference is used as the first alternative measure of the disposition effect.

One weakness of our study's disposition effect measure and of the Odean (1998) measure is that they are mechanically correlated with investor turnover (Feng and Seasholes, 2005). Imagine two investors, Alex and Bob. On any given day, Alex's propensity to sell a winning stock

(PGR) is 2% and his propensity to sell a losing stock (PLR) is 1%. Suppose that the likelihood that Bob sells both his winning and losing stocks is twice the likelihood that Alex does so on any given day. In other words, Bob's PGR is 4% and PLR is 2%. Based on Equation (1), the disposition effect for Bob is stronger (2%) than that for Alex (1%).

To address this concern, we adopt an alternative method, following Seru, Shumway, and Stoffman (2010), and estimate a proportional hazard model,

$$h_{i,j}(t) = \phi_i(t) \exp(\gamma_i \text{Gain}_{i,j,t-1} + \text{controls} + \varepsilon_{i,j,t}),$$

where $h_{i,j}(t)$ is the probability that investor i sells position j on date t , conditional on not having sold that position prior to date t . $\text{Gain}_{i,j,t-1}$ is defined in the same way as in Equation (1). Therefore, $\exp(\gamma_i)$ measures the PGR/PLR. Moreover, following Seru, Shumway, and Stoffman (2010), we add three additional controls—five-day moving averages of market returns, squared market returns, and market trading volume—to our Cox regression to control for market-driven confounding factors that may contaminate our results.

B. Lottery preference

In the paper, we construct our lottery measure based on three stock characteristics: price per share, idiosyncratic volatility, and idiosyncratic skewness. We use these three characteristics as alternative measures of lottery preferences.

C. Extrapolation

In the paper, we measure extrapolation as returns in excess of the market over the past 126 trading days. As an alternative measure, we calculate the extrapolation measure over the preceding 252 trading days.

D. Underdiversification

In the paper, we measure underdiversification as the negative of the natural logarithm of the number of stocks in a given account. If an investor holds multiple simulated accounts, we calculate the average of the number of stocks across those simulated accounts before we take the natural logarithm. With the alternative measure we treat all the simulated accounts as one and calculate the total number of stocks from all the simulated accounts.

E. Local bias

Alternatively, we measure local bias using the excess weight with which an investor buys stocks in his/her own province or prefecture cities. The benchmarks are the market capitalization of local stocks (for firms located in either the same province or the same prefecture city) as a fraction of the market capitalization of all listed stocks.

F. Turnover

We do not consider any alternative measures of turnover.

G. Investor performance

In the paper, we measure investor performance over the two months following purchases. As alternative measures, we measure investor performance over the following one, three, or six months.

Section H. Pooled regressions for the disposition effect analysis

The individual-level disposition effect estimates may be noisy. As an alternative, we estimate the disposition effect by pooling all real accounts or all simulated accounts together. Specifically, we estimate the following model,

$$Sell_{i,j,t} = \beta_1 Gain_{i,j,t-1} + \beta_2 Real_{i,j} + \beta_3 Gain_{i,j,t-1} * Real_{i,j} + controls + \varepsilon_{i,j,t}$$

where observations occur at the investor (i), position (j), and date (t) levels. $Sell_{i,j,t}$ is a dummy variable that equals 1 if investor i sells position j (partially or fully) on day t . $Gain_{i,j,t-1}$ is a dummy variable that equals 1 if investor i experiences a gain on stock j at the end of day $t-1$. $Real_{i,j}$ is a dummy variable that equals 1 if position j is in a real account and 0 otherwise.

In addition to estimating a linear probability model that includes the pooled data, we also estimate a hazard model,

$$h_{i,j}(t) = \phi(t) \exp(\beta_1 Gain_{i,j,t-1} + \beta_2 Real_{i,j} + \beta_3 Gain_{i,j,t-1} * Real_{i,j} + controls + \vartheta_{i,j,t}),$$

where $h_{i,j}(t)$ is the probability that investor i sells position j on date t , conditional on not having sold that position prior to date t . Other variables are defined as they are in the linear probability model.

In Table A8 we report the linear probability model results and in Table A9 we report the hazard model results. We multiply the coefficients listed in Table A8 by 100. The results indicate that, in both specifications, β_3 is consistently positive. The results are also robust across various high-dimensional fixed-effect specifications or stratification specifications.

Section I. Investors holding only one simulated account

Among the sample investors, 3,332 hold multiple simulated accounts. In Table A10, we report our main results based on investors who hold only one simulated account. There are 1,081 such investors. The results are qualitatively similar to those reported in the main text.

Section J. Data truncation

If an investor has traded more than 200 times in real accounts, Snowball keeps only the most recent 200 trades. For Table A11 we re-run the analysis to address this truncation issue. In Panel A, we report the results using the sample that includes only investors who have executed fewer than 200 trades. In Panel B, we report the results using data for the year before we collect our data. The results reported in both panels are qualitatively similar to the main results.

Section K. Postings—varying windows

In Table A12, we report the results for postings over varying windows. In Table 11, we measure postings within the three days following a transaction. In Table A12, we measure postings on the day following a transaction (Panel A) or within seven days following a transaction (Panel B). The results are qualitatively similar to those reported in Table 10.

Section L. Portfolio weight and the disposition effect in Snowball data

In Table A13, we examine how a position's portfolio weight is associated with the disposition effect using Snowball data. The coefficients of *Gain*PortWeight* are positive in all subsamples and statistically significantly so in three out of five cases, suggesting that the disposition effect is stronger when a position's portfolio weight is greater.

Section M. The reverse disposition effect, following Chang, Solomon, and Westerfield (2016)

In Table A14, we examine how a fund's portfolio weight is associated with the reverse disposition effect using the US brokerage data. In this table, we follow Chang, Solomon, and Westerfield (2016) and focus on investor-months with sales. The coefficient of *Gain*PortWeight* is negative, suggesting that the reverse disposition effect is stronger when portfolio weight is greater, consistent with the results reported in Table 13.

Across the five subsamples the total number of observations is 304,698. The number of observations in Chang, Solomon, and Westerfield (2016) is 354,125. Our sample is smaller mainly because we require at least two funds to be held in an investor's portfolio. Specifically, our sample is smaller for several reasons. First, following Chang, Solomon, and Westerfield (2016), we obtain a sample of 349,459 observations, a figure that is very close to that of Chang, Solomon, and Westerfield (2016). Second, we drop fund positions with negative holdings, likely because of short positions (Hartzmark, 2015). A short position is considered when its holding is negative at any point between the initiation of the position and its full liquidation. The number of observations

reduces to 338,251. Removing investor-days with the number of funds below 2 results in a sample of 306,707 observations. The investor, stock, and day fixed effects lead to 2,009 singleton observations.

Table A1. Using simulated accounts where a performance-maximization purpose is stated explicitly

In this table, we report the differences between real and simulated accounts for a subsample of investors who explicitly state their purpose as maximizing simulated performance. In Panel A, these accounts are identified manually. In Panel B, these accounts are identified if they contain a set of keywords related to performance maximization. See Section 4.1 for the bias definitions. For each bias we report the mean value of real accounts, the mean value of simulated accounts, and the mean value of the differences between them. For the differences, we report *t*-statistics and Wilcoxon *p*-values. In column “N” we report the number of investors.

Panel A. Manual identification

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	<i>t</i> -stat	Wilcoxon <i>p</i>
Disposition effect	2,454	3.904	0.593	3.311	(22.42)	0.000
Lottery preference	2,450	5.838	3.895	1.943	(8.89)	0.000
Extrapolation	2,452	14.563	12.359	2.204	(3.70)	0.952
Underdiversification	2,469	-1.118	-1.522	0.405	(24.27)	0.000
Local Bias	1,084	0.668	0.677	-0.009	(-0.27)	0.172
Turnover	2,463	5.719	1.018	4.701	(34.22)	0.000
Performance	2,455	-0.475	0.036	-0.512	(-3.23)	0.007

Panel B. Keyword-based identification

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	<i>t</i> -stat	Wilcoxon <i>p</i>
Disposition effect	2,699	3.851	0.549	3.302	(23.92)	0.000
Lottery preference	2,696	5.943	3.764	2.178	(11.04)	0.000
Extrapolation	2,697	14.612	12.111	2.501	(4.52)	0.969
Underdiversification	2,718	-1.132	-1.562	0.430	(26.88)	0.000
Local Bias	1,157	0.637	0.659	-0.022	(-0.68)	0.597
Turnover	2,709	5.672	0.908	4.764	(36.18)	0.000
Performance	2,703	-0.395	0.059	-0.454	(-3.12)	0.015

Table A2. Excluding investors who may use simulated accounts as a “watch list”

In this table, we report differences between real and simulated accounts for a subsample of investors. These investors, in our sample period, have never purchased any stocks for their real accounts and at the time of the purchase the stocks were in their simulated accounts. See Section 4.1 for the bias definitions. For each bias we report the mean value of real accounts, the mean value of simulated accounts, and the mean value of the differences between them. For the differences, we report t -statistics and Wilcoxon p -values. In column “N” we report the number of investors.

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	794	6.761	0.278	6.483	(18.20)	0.000
Lottery preference	788	9.062	4.736	4.326	(8.62)	0.000
Extrapolation	790	22.114	8.562	13.552	(9.25)	0.000
Underdiversification	798	-0.833	-1.054	0.222	(7.06)	0.000
Local Bias	278	0.576	0.625	-0.049	(-0.62)	0.509
Turnover	792	7.153	0.343	6.810	(21.23)	0.000
Performance	792	-2.000	-0.364	-1.635	(-5.19)	0.000

Table A3. Time-series design

In this table, we report differences in biases between investors' real accounts and simulated accounts before an investor opens a real account. See Section 4.1 for the bias definitions. For each bias we report the mean value of real accounts, the mean value of simulated accounts, and the mean value of the differences between them. For the differences, we report t -statistics and Wilcoxon p -values. In column "N" we report the number of investors.

	N	Overlapping Real: After adding real accounts	Non-Overlapping Simulated: Before adding real accounts	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	2,491	4.475	0.573	3.902	(24.44)	0.000
Lottery preference	2,422	6.174	4.466	1.708	(7.56)	0.000
Extrapolation	2,429	14.568	16.155	-1.587	(-2.27)	0.208
Underdiversification	2,505	-1.090	-1.411	0.321	(20.27)	0.000
Local Bias	1,068	0.626	0.617	0.009	(0.27)	0.005
Turnover	2,443	6.762	2.326	4.435	(25.79)	0.000
Performance	2,435	-0.496	0.369	-0.865	(-5.37)	0.001

Table A4. By the ratio of overlapped holdings

In this table, we report differences in biases between real and simulated accounts for the subsample of investors with varying levels of overlapped holdings. For Panel A, we focus on investors with no overlapped holdings. For Panel B, we focus on investors with low overlapped holdings. For Panel C, we focus on investors with high overlapped holdings. See Section 4.1 for the bias definitions. For each bias, we report the mean value of real accounts, the mean value of simulated accounts, and the mean value of the differences between them. For the differences, we report t -statistics and Wilcoxon p -values. In column “N” we report the number of investors.

Panel A. No Overlapped Holdings

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	625	6.736	0.121	6.615	(16.84)	0.000
Lottery preference	618	9.127	4.307	4.821	(8.08)	0.000
Extrapolation	621	23.112	5.958	17.154	(10.08)	0.000
Underdiversification	637	-0.816	-1.028	0.213	(5.85)	0.000
Local Bias	214	0.573	0.641	-0.069	(-0.69)	0.374
Turnover	631	7.239	0.231	7.008	(19.06)	0.000
Performance	623	-2.206	-0.270	-1.936	(-5.18)	0.000

Panel B. Low Overlapped Holdings

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	1,876	4.729	0.540	4.189	(23.44)	0.000
Lottery preference	1,876	7.714	5.060	2.653	(10.19)	0.000
Extrapolation	1,877	17.214	15.060	2.154	(2.96)	0.034
Underdiversification	1,879	-1.114	-1.378	0.264	(14.46)	0.000
Local Bias	718	0.593	0.668	-0.074	(-2.06)	0.628
Turnover	1,878	6.959	0.785	6.174	(34.88)	0.000
Performance	1,877	-1.034	-0.648	-0.386	(-2.49)	0.097

Panel C. High Overlapped Holdings

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	1,870	2.736	0.562	2.174	(16.26)	0.000
Lottery preference	1,875	4.400	3.453	0.947	(5.37)	0.388
Extrapolation	1,875	11.494	12.477	-0.983	(-2.07)	0.006
Underdiversification	1,878	-1.151	-1.718	0.567	(35.96)	0.000
Local Bias	807	0.677	0.625	0.052	(1.82)	0.725
Turnover	1,877	4.531	1.098	3.433	(28.73)	0.000
Performance	1,875	0.237	0.394	-0.157	(-1.19)	0.460

Table A5. Between-subject design

In this table, we report differences in biases between investors' real and simulated accounts. For real accounts we focus on users who hold only one real account and hold no simulated accounts. For simulated accounts we use the period before an investor opens his/her real account. We require the accounts to have been held for at least one year. Another difference between this table and Table 6 is that, for Table 6, we conduct paired statistical tests of the differences, while for this table the tests are not paired.

	Real		Simulated		Difference		
	N	Mean	N	Mean	Mean	<i>t</i> -stat	Wilcoxon <i>p</i>
Disposition effect	3,289	5.435	911	0.713	4.722	(26.44)	0.000
Lottery preference	3,299	9.727	908	5.333	4.394	(12.50)	0.000
Extrapolation	3,300	20.701	909	17.551	3.149	(2.68)	0.000
Underdiversification	3,302	-1.005	911	-1.424	0.419	(17.52)	0.000
Local Bias	591	0.673	415	0.608	0.065	(1.16)	0.973
Turnover	3,301	6.818	909	1.935	4.882	(25.38)	0.000
Performance	3,300	-1.371	909	0.050	-1.421	(-6.56)	0.000

Table A6. Peer effect: Investors without any followers

In this table, we report differences between real and simulated accounts for the subsample of investors who do not have any followers. See Section 4.1 for the bias definitions. For each bias we report the mean value of real accounts, the mean value of simulated accounts, and the mean value of the differences between them. For the differences, we report t -statistics and Wilcoxon p -values. In column “N” we report the number of investors.

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	2,147	4.637	0.421	4.216	(24.26)	0.000
Lottery preference	2,147	7.597	4.716	2.882	(11.33)	0.000
Extrapolation	2,149	16.162	11.776	4.387	(6.58)	0.000
Underdiversification	2,160	-1.042	-1.403	0.362	(21.28)	0.000
Local Bias	712	0.640	0.677	-0.036	(-0.92)	0.626
Turnover	2,155	6.107	0.676	5.431	(34.44)	0.000
Performance	2,150	-1.240	-0.335	-0.905	(-5.74)	0.000

Table A7. Alternative measures

In this table, we report differences in biases between investors' real and simulated accounts based on the alternative measures described in Internet Appendix Section G.

	N	Real Mean	Simulated Mean	Mean	Difference <i>t</i> -stat	Wilcoxon <i>p</i>
Disposition effect						
Disposition effect (Odean)	2,615	8.059	1.061	6.997	(10.56)	0.000
Disposition effect (Hazard)	2,098	0.771	0.175	0.948	(3.83)	0.000
Lottery preference						
−Log(Price)	4,369	-2.793	-2.923	0.131	(17.08)	0.000
Idiosyncratic volatility	4,369	0.020	0.019	0.001	(12.29)	0.000
Idiosyncratic skewness	4,369	0.604	0.517	0.087	(13.30)	0.000
Extrapolation						
Extrapolation (252 days)	4,373	26.631	24.461	2.170	(3.55)	0.238
Underdiversification						
−Log(Number of stocks all simulated accounts)	4,394	-1.087	-2.183	1.097	(63.46)	0.000
Local Bias						
Adjusted same province ratio	1,748	0.014	0.011	0.003	(0.72)	0.109
Adjusted same prefecture ratio	1,695	-0.006	-0.005	-0.000	(-0.10)	0.145
Performance						
Performance (1m)	4,375	-0.532	-0.098	-0.434	(-5.73)	0.000
Performance (3m)	4,375	-0.630	0.006	-0.636	(-4.95)	0.001
Performance (6m)	4,375	-0.111	0.496	-0.607	(-3.26)	0.016

Table A8. Pooled regressions for the disposition effect analysis – linear probability model

In this table, we report pooled regression results obtained in the disposition effect analysis based on a linear probability model,

$$Sell_{i,j,t} = \beta_1 Gain_{i,j,t-1} + \beta_2 Real_{i,j} + \beta_3 Gain_{i,j,t-1} * Real_{i,j} + controls + \varepsilon_{i,j,t},$$

where observations occur at the investor (i), position (j), and date (t) levels. $Sell_{i,j,t}$ is a dummy variable that equals 1 if investor i sells position j (partially or fully) on day t . $Gain_{i,j,t-1}$ is a dummy variable that equals 1 if investor i experiences a gain on stock j at the end of day $t-1$. $Real_{i,j}$ is a dummy variable that equals 1 if position j is held in a real account and 0 otherwise. We add three control variables, following Ben-David and Hirshleifer (2012): the log purchase price, the square root of the number of days since purchase, and stock volatility calculated using daily returns over the 250 days preceding a purchase. For ease of presentation the coefficients are multiplied by 100. We cluster standard errors at the investor, stock, and day levels. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Real	1.071*** (7.56)	1.612*** (20.88)	1.041*** (8.07)	1.061*** (7.69)	1.613*** (20.84)	1.649*** (20.82)
Gain	0.413*** (8.25)	0.545*** (7.56)	0.518*** (7.01)	0.380*** (9.33)	0.540*** (8.28)	
Gain*Real	1.019*** (6.58)	1.022*** (6.40)	1.016*** (6.43)	0.981*** (6.46)	0.963*** (6.09)	0.890*** (6.28)
In(Price)	-0.114*** (-3.82)	-0.067*** (-2.90)	-0.103 (-1.27)	-0.123*** (-4.31)	-0.023 (-0.20)	-0.168** (-2.08)
Sqrt(Time Owned)	-0.140*** (-14.66)	-0.104*** (-15.08)	-0.132*** (-15.04)	-0.147*** (-15.76)	-0.109*** (-17.31)	-0.104*** (-17.48)
Volatility	1.277*** (5.74)	0.846*** (5.62)	0.589** (2.35)	1.397*** (6.36)	0.779*** (3.70)	0.770*** (3.66)
Constant	2.587*** (10.04)					
Investor FE	No	Yes	No	No	Yes	No
Stock FE	No	No	Yes	No	Yes	Yes
Date FE	No	No	No	Yes	Yes	Yes
Investor*Gain FE	No	No	No	No	No	Yes
Adj-R ²	0.015	0.049	0.019	0.016	0.053	0.058
Obs.	31,323,672	31,323,672	31,323,667	31,323,672	31,323,667	31,323,663

Table A9. Pooled regressions for the disposition effect analysis – hazard model

In this table, we report pooled regression results obtained in the disposition effect analysis based on a Cox proportional hazard model,

$$h_{i,j}(t) = \phi(t) \exp(\beta_1 \text{Gain}_{i,j,t-1} + \beta_2 \text{Real}_{i,j} + \beta_3 \text{Gain}_{i,j,t-1} * \text{Real}_{i,j} + \text{controls} + \vartheta_{i,j,t}),$$

where observations occur at the investor (i), position (j), and date (t) levels. $h_{i,j}(t)$ is the probability that investor i sells position j on date t , conditional on not having sold that position prior to date t . $\text{Gain}_{i,j,t-1}$ is a dummy variable that equals 1 if investor i experiences a gain on stock j at the end of day $t-1$. $\text{Real}_{i,j}$ is a dummy variable that equals 1 if position j is held in a real account and 0 otherwise. We add log purchase price and volatility as control variables. We cluster standard errors at the investor level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
Gain	0.099*** (5.60)	0.151*** (10.15)	0.118*** (7.06)	0.078*** (5.16)	
Real	1.104*** (25.31)	1.182*** (41.14)	0.989*** (24.78)	0.968*** (23.79)	1.175*** (40.74)
Gain*Real	0.160*** (8.41)	0.194*** (11.59)	0.188*** (10.41)	0.169*** (9.33)	0.186*** (9.68)
In(Purchase Price)	-0.056*** (-6.39)	-0.056*** (-9.83)	-0.046* (-1.88)	-0.056*** (-7.18)	-0.054*** (-9.42)
Volatility	0.151*** (6.77)	0.084*** (3.50)	-1.037*** (-14.48)	0.206*** (9.13)	0.087*** (3.18)
Investor stratification	No	Yes	No	No	No
Stock stratification	No	No	Yes	No	No
Date stratification	No	No	No	Yes	No
Investor*Gain stratification	No	No	No	No	Yes
Pseudo-R ²	0.013	0.032	0.029	0.022	0.034
Obs.	25,278,983	25,278,983	25,278,983	25,278,983	25,278,983

Table A10. Investors with one simulated account

In this table, we report differences between real and simulated accounts for the subsample of investors who hold only one simulated account. See Section 4.1 for the bias definitions. For each bias we report the mean value of real accounts, the mean value of simulated accounts, and the mean value of the differences between them. For the differences, we report t -statistics and Wilcoxon p -values. In column “N” we report the number of investors.

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	1,071	4.574	0.372	4.202	(17.78)	0.000
Lottery preference	1,067	7.085	3.615	3.470	(9.34)	0.000
Extrapolation	1,069	16.385	10.949	5.436	(5.24)	0.000
Underdiversification	1,081	-1.032	-1.234	0.203	(8.17)	0.000
Local Bias	353	0.661	0.696	-0.035	(-0.55)	0.166
Turnover	1,074	5.941	0.616	5.326	(23.48)	0.000
Performance	1,071	-1.068	0.155	-1.223	(-4.83)	0.001

Table A11. Data truncation issue

In this table, we report differences between real and simulated accounts for the subsample of investors who execute fewer than 200 trades in their real accounts (Panel A) and the subsample of investors who were active in the last year of our sample period (Panel B). See Section 4.1 for the bias definitions. For each bias we report the mean value of real accounts, the mean value of simulated accounts, and the mean value of the differences between them. For the differences, we report t -statistics and Wilcoxon p -values. In column “N” we report the number of investors.

Panel A. Investors who execute fewer than 200 trades in their real accounts

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	3,540	3.780	0.470	3.309	(27.56)	0.000
Lottery preference	3,540	6.430	4.361	2.069	(11.45)	0.000
Extrapolation	3,543	14.722	12.657	2.066	(4.15)	0.330
Underdiversification	3,561	-1.051	-1.461	0.410	(31.23)	0.000
Local Bias	1,421	0.621	0.640	-0.019	(-0.73)	0.671
Turnover	3,555	5.078	0.817	4.262	(41.24)	0.000
Performance	3,544	-0.551	-0.112	-0.439	(-3.87)	0.008

Panel B. Investors who are active in the last year of our sample period

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	t -stat	Wilcoxon p
Disposition effect	1,584	4.173	0.286	3.887	(18.67)	0.000
Lottery preference	1,566	4.273	2.421	1.853	(6.87)	0.000
Extrapolation	1,567	8.339	6.400	1.939	(3.27)	0.106
Underdiversification	1,610	-1.250	-1.562	0.313	(15.43)	0.000
Local Bias	640	0.638	0.640	-0.002	(-0.05)	0.286
Turnover	1,589	4.498	0.581	3.917	(24.14)	0.000
Performance	1,570	-1.144	-0.204	-0.924	(-4.15)	0.003

Table A12. Postings – one-day window or seven-day window

In this table, we report the likelihood (in %) that an investor posts about his/her transaction on Snowball on the day following the transaction (Panel A) or within seven days following the transaction (Panel B). Each row displays results for a specific type of transaction: buying, selling, selling winners, and selling losers. In the last row we also report differences between selling winners and selling losers. The analyses are conducted separately for real and simulated accounts. For each type of transaction we report the mean values of real accounts, the mean values of simulated accounts, and the mean values of the differences between them. For the differences, we report *t*-statistics and Wilcoxon *p*-values. In column “N” we report the number of investors.

Panel A. One-day window

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	<i>t</i> -stat	Wilcoxon <i>p</i>
Buying	4,409	2.731	0.776	1.959	(19.07)	0.000
Selling	4,400	1.860	0.676	1.370	(10.69)	0.000
Selling winners	4,337	2.337	0.771	1.811	(11.20)	0.000
Selling losers	4,306	1.411	0.435	1.216	(9.03)	0.000
Selling winners – Selling losers	4,243	0.926	0.381	0.560	(3.71)	0.000

Panel B. Seven-day window

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	<i>t</i> -stat	Wilcoxon <i>p</i>
Buying	4,409	4.896	1.620	3.287	(23.35)	0.000
Selling	4,400	3.097	1.470	1.939	(11.62)	0.000
Selling winners	4,337	3.932	1.869	2.564	(11.56)	0.000
Selling losers	4,306	2.222	0.998	1.546	(9.02)	0.000
Selling winners – Selling losers	4,243	1.685	0.871	0.991	(4.46)	0.000

Table A13. The disposition effect and portfolio weights – Snowball data

In this table, we report the pooled regression results obtained in the disposition effect analysis based on a linear probability model by the number of stocks in an investor's portfolio,

$$Sell_{i,j,t} = \beta_1 Gain_{i,j,t-1} + \beta_2 PortWeight_{i,j} + \beta_3 Gain_{i,j,t-1} * PortWeight_{i,j,t-1} + controls + \varepsilon_{i,j,t},$$

where observations occur at the investor (i), position (j), and date (t) levels. $Sell_{i,j,t}$ is a dummy variable that equals 1 if investor i sells position j (partially or fully) on day t . $Gain_{i,j,t-1}$ is a dummy variable that equals 1 if investor i experiences a gain on stock j at the end of day $t-1$. $PortWeight_{i,j,t-1}$ is the original weight of position j in the account when investor i established position j . We add three control variables, following Ben-David and Hirshleifer (2012): the log purchase price, the square root of the number of days since purchase, and stock volatility calculated using daily returns during the 250 days preceding a purchase. For ease of presentation the coefficients are multiplied by 100. We cluster standard errors at the investor, stock, and day levels. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1) #Stocks=2	(2) #Stocks=3	(3) #Stocks=4	(4) #Stocks=5	(5) #Stocks>5
Gain	1.964*** (12.28)	1.547*** (10.97)	1.449*** (10.59)	1.205*** (7.62)	0.926*** (10.39)
PortWeight	1.245** (1.98)	1.781** (2.42)	2.952*** (2.91)	0.354 (0.33)	1.511** (2.14)
Gain*PortWeight	0.712 (1.54)	1.573** (2.21)	0.165 (0.19)	2.161* (1.87)	2.055** (2.10)
Log(Price)	1.562*** (3.79)	-0.260 (-0.75)	0.276 (0.84)	-0.080 (-0.21)	-0.039 (-0.26)
Sqrt(Time owned)	-0.090*** (-4.64)	-0.080*** (-4.63)	-0.006 (-0.35)	0.055*** (2.63)	-0.037*** (-4.13)
Volatility	0.057 (0.47)	0.992 (0.94)	-0.679 (-1.55)	1.971 (1.46)	-0.075 (-0.18)
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Adj-R ²	0.138	0.119	0.102	0.094	0.058
Obs.	765,625	726,261	600,384	462,106	1,309,642

Table A14. The reverse disposition effect on funds based on the US brokerage data set – investor-months with sales

This table presents pooled regression results obtained in the reverse disposition effect analysis based on the equity mutual fund sample. Specifically, we report regression results of the following model based on the number of funds in an investor's portfolio,

$$Sell_{i,j,t} = \beta_1 Gain_{i,j,t-1} + \beta_2 PortWeight_{i,j,t-1} + \beta_3 Gain_{i,j,t-1} * PortWeight_{i,j,t-1} + controls + \varepsilon_{i,j,t}$$

where observations occur at the investor (i), position (j), and date (t) levels. $Sell_{i,j,t}$ is a dummy variable that equals 1 if investor i sells position j (partially or fully) on day t . $Gain_{i,j,t-1}$ is a dummy variable that equals 1 if investor i experiences a gain on fund j at the end of day $t-1$. $PortWeight_{i,j,t-1}$ is the portfolio weight of position j in investor i 's portfolio on day $t-1$. We measure the value of a position using the purchase price of a stock instead of the current price. We add two control variables, following Ben-David and Hirshleifer (2012): the log purchase price and the square root of the number of days since purchase. For ease of presentation the coefficients are multiplied by 100. We cluster standard errors at the investor, stock, and day levels. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1) #Funds=2	(2) #Funds=3	(3) #Funds=4	(4) #Funds=5	(5) #Funds>5
Gain	-0.004 (-0.32)	-0.014 (-1.13)	-0.007 (-0.64)	-0.015 (-1.14)	-0.017*** (-3.03)
PortWeight	-0.023 (-0.62)	-0.185*** (-4.35)	-0.226*** (-4.19)	-0.273*** (-3.80)	-0.288*** (-5.90)
Gain*PortWeight	-0.081** (-2.16)	-0.131*** (-2.73)	-0.249*** (-4.51)	-0.263*** (-3.44)	-0.275*** (-6.00)
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
Adj-R ²	0.384	0.341	0.326	0.318	0.229
Obs.	37,376	39,213	36,989	32,022	159,098